Fractional evolution, gamma quotients, and normalized Taylor remainders

Research manuscript prepared with ChatGPT for Keio

3 November 2026

Abstract

Initial states for a fractional Schrödinger equation can give identical terminal observations at exceptional orders and times. We construct two infinite families of such exceptions, one for a scalar multiplier and one for the determinant governing two initial data. Each has a separate finite counterexample certified by interval arithmetic. At orders at most one half a Gaussian scale-mixture representation gives logarithmic interpolation of the solution norm. For a different fractional equation we certify the order at which the first positive zero appears, its square-root splitting, and its drift at the integer endpoint.

Two gamma-quotient questions lead to a comparison of Laplace densities and a sharp threshold for a reciprocal. For normalized Taylor remainders we prove sign theorems for squared tangent and arctangent, determine optimal constants for products indexed by majorized vectors, and show that raising the arcsine quotient to a large power destroys logarithmic absolute monotonicity at every fixed truncation. A final graph theorem gives a sharp linear leaf bound. Appendices treat typical shifts in Erdős–Graham Problem 461 and classify periodic free-isospectral potentials on at most four occupied sites.

How to read the claims. The ten numbered entries on the next page are conjectural statements posed in the cited sources. We give mathematical proofs or disproofs in their original ranges. They share techniques, and their ages and difficulty vary. The number counts original statements, not distinct historically unprecedented methods. The source comparisons below identify known ingredients, alternative proofs, and specific priority uncertainties. External mathematical review and historical verification remain open.

1 Ten source conjectures and their outcomes

The page references lead to the proofs. In entries 4–5, LCM means logarithmic complete monotonicity. Two separately published questions about reciprocal tangent quotients, entries 6–7, share one proof. Entry 8 includes three related inequalities in the authors' Fifth Conjecture and is counted once.

	Original statement	Result and scope	Page
1	Chorfi et al., Conj. 2 (2025)	False. Infinitely many singular terminal-data determinants, with order asymptotics and a certified finite example.	5
2	Chorfi et al., Conj. 3 (2025)	True. A probability mixture proves complete monotonicity for every $0 < a \leq 1/2$ and the proposed norm interpolation.	7
3	Chaudhary et al., Conj. 2 (2024)	False as written. The endpoint coefficient exceeds 1.01. The threshold, strict monotonicity and square-root law hold.	11
4	Guo–Qi, Remark 4 (2011)	True. The normalized gamma-ratio derivative or its reciprocal is LCM in the respective full shift ranges.	14
5	Qi–Wei–Guo, Remark 4.5 (2012)	False. An explicit counterexample and the sharp reciprocal LCM threshold for every shift.	16
6	Liu–Qi, Remark 4.1 (2025)	True. Strict concavity of the higher divided by the lower squared-tangent remainder on $(-\pi/2, \pi/2)$.	17
7	Shuang–He–Qi, Guess 2 (2025)	True. All derivative signs of the reciprocal tangent quotient on both half-intervals.	17
8	Qi, Fifth Conjecture (2024–26 sources)	True. A comparison on constant-sum beta slices also gives both sharp constants in He–Qi Problem 3.	19
9	Qi, Sixth Conjecture (2025)	False. Exact counterexample at $q = 56$ and opposite eventual behavior in the two parameter directions.	22
10	Liu–Qi, Seventh Conjecture in Qi's survey (2026)	True. Strict log-concavity of every normalized arctangent tail. The common upper endpoint 1 is sharp.	23

Further exact conclusions and their status. Chorfi et al.'s Conjecture 1 is false. Section 2 gives infinitely many imaginary-axis crossings and a certified enclosure. Hilfer and Seybold had already plotted zero trajectories indicating the crossings in 2006. For the 2005 Graffiti.pc Conjecture 165, Section 15 gives a sharp spanning-tree bound. A 2006 conference abstract by Foster prevents a firm priority judgment. Section 16 solves a two-parameter open classification question of Qi and Luo.

Chaudhary et al.'s Conjectures 1 and 3 contain exact inequalities and numerical fits with no specified tolerance. Section 5 proves the inequalities and gives exact asymptotic information in place of the fits. Section 13 refutes both parts of Shuang–He–Qi's joint Guess 3. Together with Section 14 the remainder sections cover the nine statements listed in Section 12.7.1 of Qi's survey arXiv:2512.15797. Two are accessible from older theorems, as explained in Section 14.

Corpus and proof design. We consulted the Verifiable Mathematical Reasoning Corpus, revision `ecd87de3348dc`. The restricted leaf estimate in [066] suggests asking how a tree assembled from components pays for neighbors of maximum-degree vertices, which yields the general graph result. The proof-transfer examples in [067] motivate separate auxiliary questions and explicit maps back to the original variables. The sparse Floquet studies [017,061] enter Appendix B. Each main proof specifies its own analytical or combinatorial inputs. The earlier ten-entry manuscript is separate.

Two transfers. For low fractional orders, the auxiliary problem is to describe the difference of two independent positive half-normal scale mixtures. Its solution produces the probability mixture for an evolution multiplier, then a norm bound for its Hilbert-space equation. For orders near two, a decaying second residue is invisible in a fixed-order asymptotic. Letting order and time vary independently balances that residue with the algebraic part of a determinant. A winding argument then proves loss of uniqueness for the original two initial data.

Computation. Four executable interval verifiers are included after the end of the typeset material in the single TeX file. The two nonvanishing conjectures have independent analytic disproofs and finite interval certificates. The fractional positivity and fold bounds use their indicated analytic estimates and rigorous interval calculations.

2 Imaginary-axis crossings near the exponential order

Chorfi, Et-tahri, Maniar and Yamamoto, *Forward and backward problems for abstract time-fractional Schrödinger equations*, Communications in Nonlinear Science and Numerical Simulation 157 (2026), 109758, arXiv:2510.03600, Section 5, Conjecture 1, ask whether $E_{a,1}(ix)$ is nonzero for every real x and every $3/5 < a < 1$. Here $E_{a,b}(z) = \sum_{j\ge 0} z^j/\Gamma(aj+b)$. A phase variable and an independent damping variable give zeros approaching the order 1.

Theorem 2.1. *For every sufficiently large positive integer k there are $3/5 < a_k < 1$ and $x_k > 0$ with $E_{a_k,1}(ix_k) = 0$. They may be chosen so that $x_k \sim 2\pi k$ and*

$$\pi^2 k(1-a_k) = 2\log k - \log\log k + 3\log\pi + o(1).$$

In particular infinitely many distinct orders give counterexamples.

Proof. Let $a = 1-d$, $0 < d \le 1/10$, and $z = iX$ with $X \ge 1$. Deformation of the inverse Laplace contour for $E_{a,1}(zu^a)$ at $u = 1$ gives

$$E_{a,1}(z) = a^{-1}e^{z^{1/a}} + J_a(z), \qquad J_a(z) = -\frac{z\sin(\pi a)}{\pi}\int_0^\infty \frac{e^{-r}r^{a-1}}{(r^a e^{i\pi a} - z)(r^a e^{-i\pi a} - z)}\,dr.$$

Indeed the transform $s^{a-1}/(s^a - z)$ on the principal sheet has one simple pole at $z^{1/a}$ of residue $1/a$. The two sides of the negative real cut give the displayed difference. An inner arc has integral $O(\epsilon^a)$, and on a left semicircle of large radius R the transform is $O(R^{-1})$. The factor e^s makes its arc integral tend to zero by the usual Jordan estimate. One may start on a line to the right of all singularities and close about its intersection with the real axis.

The angular separations imply $|r^a e^{\pm i\pi a} - iX| \ge c(r^a + X)$ for an absolute $c > 0$. The reciprocal of the product differs from $1/z^2$ by at most Cr^a/X^3. This follows by subtracting the fractions and bounding the resulting numerator by $2Xr^a + r^{2a}$. Integration and Euler's reflection formula now give, uniformly in this range,

$$J_a(iX) = \frac{i}{X\Gamma(1-a)} + O(d/X^2) = \frac{i}{X\Gamma(d)}(1 + O(X^{-1})).$$

In particular the algebraic term stays in a fixed sector around the positive imaginary axis, even when d approaches zero.

Put $\phi = \pi/(2a)$, use coordinates (d,θ) and let $t = \theta/\sin\phi$, $X = t^a$. On the rectangle

$$k^{-2} \le d \le \log(k)/k, \qquad 2\pi k + \pi \le \theta \le 2\pi k + 2\pi,$$

one has $t \sim 2\pi k$ and $X \sim 2\pi k$ uniformly. Define the real continuous functions

$$U = t\cos\phi - \log(a|J_a(iX)|), \qquad V = \theta - 2\pi k - \arg[-J_a(iX)],$$

taking the branch of argument near $3\pi/2$. On the two θ faces V has opposite strict signs. On the lower d face, $t\cos\phi = O(k^{-1})$ and $a|J_a(iX)| \asymp k^{-3}$, so $U > 0$. On the upper face, $t\cos\phi = -(\pi^2 + o(1))\log k$ and $\log(a|J_a(iX)|) = -2\log k + O(\log\log k)$, so $U < 0$. The two-dimensional intermediate value theorem (Poincaré–Miranda) gives a common zero. At it $e^{(iX)^{1/a}} = -aJ_a(iX)$, proving the required cancellation.

Throughout this rectangle $t\cos\phi = -\pi^2 kd + o(1)$ because $kd^2 = o(1)$, and $\log(a|J_a(iX)|) = \log d - \log(2\pi k) + o(1)$. At a common zero $w = \pi^2 kd$ therefore satisfies $w + \log w = 2\log k + \log(2\pi^3) + o(1)$. First $w \sim 2\log k$, then substitution gives the stated expansion. Since $d > 0$ tends to zero, the orders cannot form a finite set. □

On a scalar eigenmode with eigenvalue $\lambda > 0$, the time-fractional equation $i\partial_t^a u - \lambda u = 0$ (Caputo derivative) has $u(T) = E_{a,1}(-i\lambda T^a)u(0)$, as follows by a Laplace transform. By complex conjugation, for each crossing above, $T = (x_k/\lambda)^{1/a_k}$ gives identical terminal data for every initial amplitude. Thus an all-time uniqueness claim for the backward problem already fails on a single eigenmode.

Earlier zero trajectories. R. Hilfer and H. J. Seybold, *Computation of the generalized Mittag–Leffler function and its inverse in the complex plane*, Integral Transforms and Special Functions 17 (2006), 637–652, Figure 1 and Theorems 3.1–3.2, plotted zero trajectories across the orders and studied endpoint limits for fixed-index zeros. Their plotted paths already indicate imaginary-axis crossings. This prior work is material to the 2025 conjecture. The preceding direct proof gives a large-index law for the crossing orders and the resulting nonuniqueness times. Historical priority for the qualitative observation is not asserted.

A finite counterexample enclosure. For an independent verification of the zero, set $F(a,t) = (\Re E_{a,1}(it), \Im E_{a,1}(it))$,

$$x_0 = (0.74785642795992934307910824 26,\ 3.71313849637937052367138331 85),$$

$$B = 10^{-3}\begin{pmatrix} 424 & -523 \\ 8133 & 792 \end{pmatrix}.$$

On $D = x_0 + [-10^{-12}, 10^{-12}]^2$, outward-rounded interval evaluation gives

$$\|BF(x_0)\|_\infty < 3 \cdot 10^{-29}, \qquad \sup_{x\in D} \|I - BDF(x)\|_\infty < 0.000132.$$

Thus $x \mapsto x - BF(x)$ contracts D into itself. Since B is invertible, its fixed point gives a zero in D.

These two bounds can be reproduced by evaluating the first 240 terms of the complex series at 160 bits of Arb precision, multiplying each summand $v_k = (it)^k/\Gamma(1+ak)$ by $-k\psi(1+ak)$ and k/t for the two derivative columns, respectively. For $k \geq 240$ each resulting summand is bounded in modulus by $(k+1)^2 4^k/\lfloor k/2 \rfloor!$, since $1/2 < a < 1$, $1 < t < 4$ and $0 < \psi(1+ak) < \log(1+ak)$. For $k = 2m, 2m+1$ together the bound is $5(2m+2)^2 4^{2m}/m!$. Consecutive such bounds have ratio below $1/4$ for $m \geq 120$. Thus a radius 10^{-42} for each real or imaginary tail suffices.

3 Zeros of the backward evolution determinant

Conjecture 2 in Section 5 of Chorfi et al. concerns the determinant in their equation (4.14). With $\lambda = i^{-a} = e^{-i\pi a/2}$, it is

$$\Psi_a(t) = E_{a,1}(\lambda t)^2 - \lambda t E_{a,2}(\lambda t) E_{a,a}(\lambda t).$$

They conjecture $\Psi_a(t) \neq 0$ for every $t > 0$ and $1 < a < 2$. The one-pole expansion for fixed order in their Lemma 4.10 leaves out an exponentially small contribution. As a approaches 2 that contribution can cancel the algebraic term.

Theorem 3.1. *Let $N_k = 2\pi k + 3\pi/2$. For all sufficiently large positive integers k there exist $a_k \in (1,2)$ and $w_k = N_k + o(1)$ with*

$$\Psi_{a_k}(w_k^{a_k}) = 0, \qquad \pi^2 k(2-a_k) = 2\log k - \log\log k + \log(2\pi^3) + o(1).$$

Proof. Write $d = 2 - a$, $y_b(w) = w^{b-1} E_{a,b}(\lambda w^a)$ for $b = 1, 2$. Termwise differentiation gives $y_2' = y_1$ and $y_1' = \lambda w^{a-1} E_{a,a}(\lambda w^a)$. Thus the original determinant is $y_1^2 - y_1' y_2$ at $t = w^a$.

On $1.9 \leq a < 2$ the two simple principal-sheet poles of the transforms $s^{a-b}/(s^a - \lambda)$ are

$$\xi_0 = -i, \qquad \xi_1 = -\sin\theta + i\cos\theta, qquad\theta = \pi d/a.$$

Set $e_j = e^{\xi_j w}$. For $w \geq 1$ one has uniformly

$$\begin{aligned} y_1 &= (e_0 + e_1)/a + O(w^{-a}), \\ y_1' &= (\xi_0 e_0 + \xi_1 e_1)/a + O(w^{-a-1}), \\ y_2 &= (e_0/\xi_0 + e_1/\xi_1)/a - \frac{w^{1-a}}{\lambda\Gamma(2-a)} + O(w^{1-2a}). \end{aligned}$$

For the bounds deform the inverse Laplace contour about the negative real cut and take the two residues. On its two sides $s = re^{\pm i\pi}$ one has $|s^a - \lambda| \geq c(1 + r^a)$ with an absolute constant $c > 0$ by angular separation. The y_1 cut is bounded by a constant times $\int_0^\infty e^{-wr} r^{a-1}\, dr = O(w^{-a})$. Differentiation there inserts one factor $-r$ and proves the derivative estimate. For y_2 subtract $-s^{a-2}/\lambda$ from the cut integrand. The remaining integrand before multiplication by e^{sw} is bounded by $Cr^{2a-2}/(1 + r^a)$. The inverse cut of the subtracted power is $-w^{1-a}/(\lambda\Gamma(2-a))$ by the gamma integral. The inner arcs vanish as $a - 2 > -1$. The outer arcs vanish by exponential decay away from the ends and the $O(1/|s|)$ or stronger decay of the original transforms. All constants are bounded uniformly over the indicated orders.

The individual squared exponentials in $y_1^2 - y_1' y_2$ cancel. Define

$$C_a = \frac{2 - \xi_0/\xi_1 - \xi_1/\xi_0}{a^2} = \frac{2 + 2\cos\theta}{a^2}, \qquad B_a = \frac{\xi_0}{a\lambda\Gamma(d)}.$$

As $d \downarrow 0$, $C_a \to 1$ and $B_a \sim id/2$. Whenever $w \to \infty$, $d \to 0$, $dw \to \infty$ and $|e_1| = O(dw^{1-a})$, the preceding estimates give

$$e^{iw}\Psi_a(w^a) = C_a e_1 + B_a w^{1-a} + o(dw^{1-a}).$$

Indeed, the total remaining error is $O(w^{-a} + w^{1-2a} + |e_1| dw^{1-a})$. These bounds also give uniform little-oh estimates on the windows used below.

For $N = N_k$ choose d_N between $2\log N/(\pi N)$ and $6\log N/(\pi N)$ so that, with $a_N = 2 - d_N$,

$$C_{a_N} \exp[-N \sin(\pi d_N/a_N)] = |B_{a_N}| N^{1-a_N} =: D_N.$$

Such a choice follows by continuity. At the two endpoints the left side has orders $N^{-1}(1 + o(1))$ and $N^{-3}(1 + o(1))$, while the right side has order $(\log N)/N^2$ at both. Taking logarithms in the equality gives $d_N \sim 4\log N/(\pi N)$.

Use real coordinates $d = d_N - 2s/(\pi N)$, $w = N + v$. On any fixed bounded disk in (s, v) the exponential modulus in the preceding identity is multiplied by $e^{s+o(1)}$, since $[\sin(\pi d/(2-d))]' \to \pi/2$. The phase $w\cos\theta = N + v + o(1)$ because $Nd^2 = o(1)$. Also $B_a w^{1-a}/D_N \to i$ uniformly, since $d/d_N \to 1$ and $(a - a_N)\log N = o(1)$. Consequently the continuous maps satisfy

$$D_N^{-1} e^{iw}\Psi_a(w^a) \longrightarrow i(1 - e^{s+iv})$$

uniformly on this disk. On every sufficiently small circle about the origin the limit has winding number one, by its simple zero there. Uniform convergence preserves the winding number and forces an interior zero of the exact real two-parameter map. Choose a zero on a fixed closed disk of radius $1/4$ for each large k. Every limit point of these coordinates is a zero of the limit map on that disk and hence is the origin. This proves $w_k = N_k + o(1)$. At these zeros the modulus equation gives $\exp[-(\pi/2)Nd + o(1)] = (1 + o(1))d/(2N)$ because $Nd^2 = o(1)$. Thus $Q = (\pi/2)Nd$ satisfies $Q + \log Q = 2\log N + \log\pi + o(1)$. Using $Q \sim 2\log N$ and $N = 2\pi k + O(1)$ gives the claimed expansion. □

For an eigenvalue of magnitude one the scalar map from the two initial data to the terminal value and terminal derivative has determinant $\Psi_a(w^a)$ by the representation in the source. At the zeros in Theorem 3.1 it has a nontrivial kernel. Together with Theorem 2.1, this illustrates two different failures of terminal uniqueness as the order and observation time vary. For any fixed order the source already proves finiteness of the exceptional positive arguments (Lemma 4.11). The infinitude here ranges across orders.

A certified finite counterexample. In the coordinates (a, w) with original argument $t = w^a$, the determinant has a zero in the closed square of radius 10^{-12} about

$$(1.68430936268566911574800653125,\ 5.40314724546132388337747818913).$$

The corresponding original argument is approximately 17.13972580. Here is a separate certificate, independent of the large-index argument. Put $L = \log w - i\pi/2$, $z = e^{aL}$, and $D = E_{a,1}(z)^2 - zE_{a,2}(z)E_{a,a}(z)$. For $b \in \{1, 2, a\}$ the term $z^j/\Gamma(aj + b)$ has first derivative multipliers $jL - (j + b')\psi(aj+b)$ and aj/w in a and w, respectively, where $b' = db/da$. The program `schrodinger2.py` embedded in the source sums through $j = 127$. For $j \geq 128$, all three series tails for each b are bounded using $3(j + 1)^2 20^j/j!$ termwise. Indeed $|z| < 20$, $|L| < 4$, $\Gamma(aj + b) \geq j!$ and $0 < \psi(aj + b) < \log(2j + 2) < j$. The bounds have successive ratio below $1/2$ and their total is less than 10^{-36}.

For $P = (\Re D, \Im D)$ the fixed matrix $B = 10^{-3} \left(\begin{smallmatrix} 896 & -459 \\ 10189 & 14195 \end{smallmatrix}\right)$ is invertible. Arb enclosures at 176 bits verify that $\|BP(a_0, w_0)\|_\infty < 10^{-15}$ at the center and that $\|I - BP'\|_\infty < 1/100$ throughout the square. Thus $x \mapsto x - BP(x)$ maps the square into itself and contracts there. Its fixed point is a zero of the original determinant.

4 A completely monotone squared Mittag–Leffler modulus

Chorfi, Et-Tahri, Maniar and Yamamoto, *Forward and backward problems for abstract time-fractional Schrödinger equations*, arXiv:2510.03600, Section 5, Conjecture 3, ask whether

$$f_\alpha(t) = E_{2\alpha,1}(-t)^2 + tE_{2\alpha,1+\alpha}(-t)^2$$

is completely monotone for $0 < \alpha \leq 1/2$. Here $E_{\alpha,\beta}(z) = \sum_{k\geq 0} z^k/\Gamma(\beta + \alpha k)$. We prove the stronger representation of f_α as the Laplace transform of a probability supported on $(0, \infty)$.

An auxiliary problem. Let $X = \sqrt{A}\,|G|$, where G is centered Gaussian of variance 2 and A is any independent random variable supported on $(0, \infty)$. For an independent copy Y, is $X - Y$ a mixture of centered Gaussian laws? The answer is affirmative, including when the mixing law has atoms or has no moments.

Proof. For scale a the positive-axis density is

$$h_a(x) = (\pi a)^{-1/2} \exp(-x^2/(4a)).$$

Fix the scales a, b of X, Y, respectively. The density of $X - Y$ at $x > 0$ is

$$p_{a,b}(x) = \int_0^\infty h_a(x + y)h_b(y)\,dy = \frac{e^{-x^2/(4(a+b))}}{\sqrt{\pi(a + b)}}\ \operatorname{erfc}\left(\frac{x}{2\sqrt{a + b}}\sqrt{\frac{b}{a}}\right).$$

This identity follows by completing the square. The conditional density at $-x$ is $p_{b,a}(x)$. Because the two scales are exchangeable, we may first symmetrize the conditional density by averaging these two expressions, then average in a, b.

For $z \geq 0$,

$$\operatorname{erfc}(\sqrt{z}) = \frac{2}{\pi}\int_0^\infty \frac{e^{-z(1+s^2)}}{1 + s^2}\,ds.$$

Indeed, the two sides take value 1 at zero and have derivative $-e^{-z}/\sqrt{\pi z}$ for $z > 0$. Hence the symmetric density is a positive mixture of $e^{-\lambda x^2}$ with $0 < \lambda < \infty$. This assertion holds first for each fixed (a, b), and then for the averaged density by Tonelli's theorem. If ν is the

resulting mixing measure, integration in x gives $\int \sqrt{\pi/\lambda}\,\nu(d\lambda) = 1$. Thus $\tau(d\lambda) = \sqrt{\pi/\lambda}\,\nu(d\lambda)$ is a probability measure. The Fourier transform of a normalized Gaussian yields

$$\mathbb{E}e^{i\xi(X-Y)} = \int_{(0,\infty)} e^{-\xi^2/(4\lambda)}\,\tau(d\lambda).$$

This proves the assertion for every mixing law under consideration. □

Application to the original conjecture. For $0 < \gamma < 1$, take a positive stable random variable S_γ with Laplace transform e^{-u^γ}. For completeness, existence follows from the Lévy–Khintchine representation

$$u^\gamma = \frac{\gamma}{\Gamma(1-\gamma)} \int_0^\infty (1 - e^{-ux}) x^{-1-\gamma}\,dx.$$

Define $M_\gamma = S_\gamma^{-\gamma}$ and $M_1 = 1$. Integrating the Laplace transform against the gamma kernel gives, for every $p > 0$,

$$\mathbb{E}M_\gamma^p = \frac{1}{\Gamma(\gamma p)} \int_0^\infty u^{\gamma p - 1} e^{-u^\gamma}\,du = \frac{\Gamma(1+p)}{\Gamma(1+\gamma p)}.$$

The final ratio also applies to $\gamma = 1$. Monotone convergence in the exponential series gives finite moment generating function $E_{\gamma,1}(z)$ for all real $z \geq 0$, hence the characteristic function is $E_{\gamma,1}(i\xi)$.

When $0 < \alpha \leq 1/2$, take $A = M_{2\alpha}$ independent of G. The gamma duplication formula gives $\mathbb{E}|G|^k = k!/\Gamma(1+k/2)$ for integer $k \geq 0$. Consequently $\sqrt{A}\,|G|$ and M_α have the same moments $k!/\Gamma(1+\alpha k)$, and the finite exponential moments identify their laws. Splitting the series into even and odd parts gives

$$E_{\alpha,1}(i\sqrt{t}) = E_{2\alpha,1}(-t) + i\sqrt{t}\,E_{2\alpha,1+\alpha}(-t).$$

Taking squared moduli and applying the auxiliary result now shows that $f_\alpha(t) = \int e^{-t/(4\lambda)}\,\tau(d\lambda)$. In particular $(-1)^n f_\alpha^{(n)}(t) > 0$ for every integer $n \geq 0$ and every $t > 0$. Cauchy–Schwarz in the integral also gives log-convexity.

Logarithmic interpolation for the evolution equation. In the Hilbert-space setting of equation (3.1) of the source, write $c_j = \langle u_0, \phi_j \rangle$ for the coefficients in the orthonormal eigenbasis, with $A\phi_j = -\lambda_j \phi_j$. Its solution formula (3.3) and Parseval's identity give

$$H(t) := \|u(t)\|^2 = \sum_j |c_j|^2 f_\alpha(\lambda_j^2 t^{2\alpha}).$$

Each term is a Laplace transform in t because $e^{-dt^{2\alpha}}$ is the Laplace transform of a positive stable law rescaled by $d^{1/(2\alpha)}$ for $d > 0$, with a point mass when $2\alpha = 1$ or $d = 0$. Summation therefore represents H as a Laplace transform of a finite measure of mass $\sum_j |c_j|^2 = \|u_0\|^2$. Hölder's inequality under this measure gives $H(t) \leq H(0)^{1-t/T} H(T)^{t/T}$ for $0 < t < T$. Taking square roots proves the authors' estimate (5.1), including its two endpoints by continuity,

$$\|u(t)\| \leq \|u(0)\|^{1-t/T} \|u(T)\|^{t/T} \qquad (0 \leq t \leq T, \quad 0 < \alpha \leq 1/2).$$

The zero initial datum gives the inequality directly.

5 Uniform positivity and first zeros in two fractional kernels

Chaudhary, Diethelm and Hashemishahraki, *On the separation of solutions to fractional differential equations of order* $\alpha \in (1,2)$, Applied Numerical Mathematics **203** (2024), 84–96, doi:10.1016/j.apnum.2024.05.020, formulate Conjectures 1 and 3 for the first positive zero $Z_\beta(\alpha)$ of $E_{\alpha,\beta}(-t^\alpha)$, with $\beta = \alpha$ and 1, respectively. The variable is t, as in their equation (17). Writing $\beta = 1 + (1-\eta)(\alpha-1)$ with $\eta \in \{0,1\}$ makes it possible to treat both problems in the same argument. Put $S_0 = 2.93785$, $S_1 = 1.559$, $T_0 = \pi$, and $T_1 = \pi/2$.

Theorem 5.1. *For* $1 \le \alpha \le 2$ *one has* $E_{\alpha,\beta}(-t^\alpha) > 0$ *on* $0 \le t \le S_\eta$. *For* $\alpha = 1+\delta \downarrow 1$ *let* $L = \log(1/\delta)$ *and now allow every* $0 \le \eta \le 1$, *including parameters varying with* δ. *Uniformly throughout this interval,*

$$Z_\beta(1+\delta) = L + 2\log L - \log(1+\eta L) + o(1). \tag{5.1}$$

In particular the correction is $2\log L + o(1)$ *when* $\beta = \alpha$, *and* $\log L + o(1)$ *when* $\beta = 1$. *At the other endpoint, for* $\eta \in \{0,1\}$,

$$Z_\beta(\alpha) = T_\eta + c_\eta(2-\alpha) + O((2-\alpha)^2),$$
$$-0.810252 < c_0 < -0.810251, \qquad -0.129292 < c_1 < -0.129291.$$

The first assertion proves the exact positivity bounds in both part (a)'s. The leading coefficient of $\log(1/\delta)$ is exactly one in both part (b)'s. The printed numerical estimates 0.81 and 0.275 therefore require correction. Because the source uses a relative error $1 + o(1)$, its additive fit parameter d is unidentifiable at that precision. The additive-error estimate here specifies the next diverging term. For part (c), the displayed intervals and the formula below specify the slopes without assigning a tolerance to the authors' approximate values.

A moving-scale inverse Laplace estimate. Let $K_{\delta,\eta}(t) = t^{\beta-1}E_{\alpha,\beta}(-t^\alpha)$. Its Laplace transform, obtained by integrating the defining series first for real $p > 1$, is

$$A_{\delta,\eta}(p) = \frac{p^{\eta\delta}}{1+p^{1+\delta}}.$$

Use the principal logarithm. For $t \ge 2$ and $0 \le \delta \le 1/4$,

$$K_{\delta,\eta}(t) = e^{-t} + \delta h_\eta(t) + O(\delta^2), \qquad \mathcal{L}h_\eta(p) = H_\eta(p) = \frac{[\eta + (\eta-1)p]\log p}{(1+p)^2}. \tag{5.2}$$

Here the error is uniform even when t tends to infinity. To prove it, invert on $p = 1/t + iy$. The Bromwich line can be shifted there from real part 2 because the transform is holomorphic in the open right half-plane and the horizontal end segments vanish. Taylor expand $A_{\delta,\eta}$ to first order in δ under this inversion. Indeed, on this line $|1 + p^{1+v}| \ge c(1+|p|^{1+v})$ uniformly for $0 \le v \le 1/4$, since $|(1+v)\arg p| \le 5\pi/8$. Direct differentiation of its quotient gives the following estimates for $B = \partial_v^2 A_{v,\eta}$. On $0 < |y| \le 1$ one has $|B| \le C(1 + |\log|y||^2)$. On $|y| \ge 1$ one has

$$|B| \le C\frac{1+\log^2(2+|y|)}{|y|}, \qquad |\partial_y B| \le C\frac{1+\log^2(2+|y|)}{|y|^2}.$$

For clarity, $B = A_{v,\eta}(\log p)^2[(\eta-U)^2 - U(1-U)]$ where $U = p^{1+v}/(1+p^{1+v})$. Both U and $1-U$ are bounded, $\partial_p U = (1+v)U(1-U)/p$, and $A_{v,\eta} = O(1/|p|)$ at infinity. These bounds justify the estimates and the integrated Taylor remainder. On the two tails an integration by parts against e^{iyt} gives a uniform bound for the inverse integral of B. On the finite interval the first estimate is integrable, and $e^{t\Re p} = e$. This proves (5.2).

To determine $h_\eta(t)$, deform the inversion contour for H_η to real part $-1/2$, splitting the contour at its cut on the negative real axis. No pole at -1 is crossed. The jump of the logarithm gives

$$h_\eta(t) = -\int_0^{1/2} e^{-rt}\frac{\eta+(1-\eta)r}{(1-r)^2}\,dr + O(e^{-t/2}).$$

The small indentation around zero has vanishing integral since $H_\eta = O(1+|\log p|)$ there. The two vertical half-lines contribute $O(e^{-t/2})$, by integration by parts away from their bounded endpoints and the decay $H'_\eta = O((1+\log|p|)/|p|^2)$. Expanding the integrand factor as $\eta+(1+\eta)r+O(r^2)$ yields

$$h_\eta(t) = -\frac{\eta}{t} - \frac{1+\eta}{t^2} + O(t^{-3}) = -\frac{1+\eta t}{t^2}(1+O(t^{-1})).$$

All constants are uniform for $0 \le \eta \le 1$, since $\eta/(1+\eta t) \le 1/t$.

Fix $v > 0$ and define $t_\pm = L + 2\log L - \log(1+\eta L) \pm v$. Then $t_\pm/L = 1 + O(\log L/L)$ and $(1+\eta t_\pm)/(1+\eta L) = 1 + O(\log L/L)$ uniformly. Therefore

$$\frac{e^{-t_\pm}t_\pm^2}{\delta(1+\eta t_\pm)} \longrightarrow e^{\mp v}$$

uniformly. When the uniform remainder in (5.2) is divided by $\delta(1+\eta t)/t^2$, it is $O(\delta t^2) = o(1)$ for $2 \le t \le t_+$. The function $e^{-t}t^2/(1+\eta t)$ is decreasing for $t > 2$. Consequently K is positive on $[M, t_-]$ and negative at t_+ once M is sufficiently large in terms of v and δ sufficiently small. On $[0, M]$ the unscaled Mittag–Leffler series tends uniformly to e^{-t} as $\delta \to 0$, by compact convergence and uniform tail domination, uniformly also in η. This confines the first zero to $[t_-, t_+]$ for all these parameters. The arbitrariness of v proves (5.1). In the transition regime $\eta L \to \tau < \infty$ it supplies the constant displacement $-\log(1+\tau)$.

The regular endpoint. Write $F(\alpha, t) = E_{\alpha,\beta}(-t^\alpha)$, $b = 2-\eta$ and $T = T_\eta$. At $\alpha = 2$ the function is $\sin(t)/t$ for $\eta = 0$ and $\cos t$ for $\eta = 1$. Its first positive zero T is simple. The defining series and its first two derivatives in either variable converge locally uniformly around $(2, T)$ by the gamma growth in its terms. The implicit function theorem and compact positivity before T ensure that the local zero is the first zero. Hence

$$c_\eta = \frac{1}{F_t(2,T)}\sum_{k\ge 0}\frac{(-1)^k T^{2k}}{\Gamma(2k+b)}\big[k\log T - (k+1-\eta)\psi(2k+b)\big].$$

Here $F_t(2,T) = -1/\pi$ or -1 respectively. For $k \ge 60$ the summand has absolute value at most $6(k+1)^2 16^k/(2k)!$ using $T < 4$ and $0 < \psi(s) < \log s$ for $s \ge 2$. The successive bounds have ratio below $1/2$. The first 60 terms with this explicit remainder give the stated enclosing intervals.

Uniform positivity certificate. Let $X = t^\alpha$, $S = S_\eta$ and $m = 1-\eta$. Define $P(\alpha, X) = \sum_{k=0}^{60}(-X)^k/\Gamma(\alpha k+\beta)$. Throughout $1 \le \alpha \le 2$ and $0 \le X \le 9$, the errors in the value, the X derivative, and the α derivative taken along $X = S^\alpha$ are each strictly smaller than 10^{-13}. Indeed, for $k \ge 61$ the corresponding three term magnitudes are at most $4(k+1)^2 9^k/k!$. For the last bound use $|k\log S - (k+m)\psi(\alpha k+\beta)| \le 4(k+1)^2$ and monotonicity of Γ above 2. The successive majorants have ratio below $1/2$.

We verify strict positivity at $X = S^\alpha$ and strict negativity of the X derivative on $[0, S^\alpha]$. The accompanying verifier starts with $A = [1, 2]$ and bisects it into dyadic intervals. On each interval it computes a ball for the midpoint value at the right edge and a ball for its derivative across A. The mean value theorem then encloses the value along the edge. For the X derivative, Horner evaluation with interval inputs covers $[0, S^{\sup A}]$ after adaptive bisection. Values of

the reciprocal gamma function above 2 and of the digamma function on the positive axis are enclosed by monotonicity and evaluation at both endpoints. Every comparison is a strict Arb ball comparison after adding the tail bounds. At 112-bit precision, the calculation succeeds on 567 final parameter intervals for $\eta = 0$ (maximum depth 13) and on 37 for $\eta = 1$ (maximum depth 8). This proves positivity over the two full continua stated in the sources.

Earlier work and interpretation. For $\eta = 1$ the leading logarithmic scale of the escaping zero appears already in Gorenflo and Mainardi's earlier asymptotic balance [5, Section 3.2, Eq. (3.36)]. The present formula provides an additive $o(1)$ estimate including the $\log L$ term. For $\eta = 0$ the zero concerns the derivative of the $\eta = 1$ relaxation kernel, and its next coefficient is twice as large. Formula (5.1) describes the transition when β moves between 1 and α with the order. The certified bounds and slopes answer individually the remaining questions in the two published conjectures. Historical priority of the full collection is subject to further comparison.

6 The drift of the zeros continuing from the sine function

Let $Z_2(\alpha)$ be the least strictly positive t for which $E_{\alpha,2}(-t^\alpha) = 0$. Conjecture 2 of Chaudhary, Diethelm and Hashemishahraki [4] includes the exact assertion

$$Z_2(2 - \varepsilon) = \pi + \varepsilon + o(\varepsilon).$$

This is part (b)(i) in the journal version and part (a)(i) in the arXiv version. We compute the drift at every positive sine zero. The first already contradicts the stated coefficient.

Theorem 6.1. *For each integer $k \geq 1$, there is a unique real zero $t_k(\alpha)$ near $T = k\pi$ for α near 2, and*

$$t_k(2 - \varepsilon) = T + \frac{T\,\mathrm{Ci}(T) + \mathrm{Si}(T)}{2}\,\varepsilon + O_k(\varepsilon^2).$$

Here $\mathrm{Si}(x) = \int_0^x (\sin v)/v\,dv$ and $\mathrm{Ci}(x) = \gamma + \log x + \int_0^x (\cos v - 1)/v\,dv$. The first branch is $Z_2(\alpha)$. Its coefficient exceeds 1.01 and is approximately 1.041686.

Proof. Put $Y(\alpha, t) = tE_{\alpha,2}(-t^\alpha)$. At $\alpha = 2$ it equals $\sin t$. The defining series and its first two derivatives in α, t converge locally uniformly for $t > 0$ and α near 2. This follows from Stirling's formula, since the derivatives add only polynomial and logarithmic factors in the summation index on each compact rectangle. The implicit function theorem at T applies because $\cos T \neq 0$. For the first zero, choose a small fixed interval about π where the derivative in t stays negative. On the complementary interval from 0 to its left endpoint, $E_{2,2}(-t^2) = \sin(t)/t$ is strictly positive with limiting value one at zero. Uniform convergence of the defining series there preserves this positivity for α close to 2.

Write $H_j = \sum_{l=1}^{j} 1/l$ and $A(t) = (t\cos t - \sin t)/2$. Termwise differentiation gives

$$\partial_\alpha Y(2, t) = \sum_{n \geq 0} \frac{(-1)^n n t^{2n+1}}{(2n+1)!}(\log t + \gamma - H_{2n+1}).$$

The same sum without the last factor equals $A(t)$. Using $H_j = \int_0^1 (1 - u^j)/(1 - u)\,du$, one obtains

$$\partial_\alpha Y(2, t) = (\log t + \gamma)A(t) - \int_0^1 \frac{A(t) - A(tu)}{1 - u}\,du.$$

Termwise integration is justified by the convergent absolute series weighted by H_{2n+1}. For $t = T$ substitute $v = T(1 - u)$. The numerator of the integral becomes

$$\frac{\cos T}{2}\{T(1 - \cos v) + v\cos v - \sin v\}.$$

Since $\int_0^T \cos v\,dv = 0$, the preceding identities give

$$\partial_\alpha Y(2,T) = \frac{\cos T}{2}\bigl(T\,\mathrm{Ci}(T) + \mathrm{Si}(T)\bigr).$$

Implicit differentiation with respect to $\varepsilon = 2-\alpha$ gives the displayed drift after division by $\cos T$.

For an elementary strict certificate set $p = 157/50$ and $q = 22/7$, so $p < \pi < q$. Convexity of $1/x$ and the trapezoidal rule on $[5,m]$ give $\gamma \ge H_5 - \log 5 - 1/10$ on letting $m \to \infty$. The first four terms of the exponential series show $e^{47/100} > 5/p$, so $\gamma + \log\pi > H_5 - 1/10 - 47/100$. For $j \ge 0$ let $r_j = p$ if j is even and $r_j = q$ if j is odd. The alternating series of the two integrals have decreasing term sizes at π (by the ratio test on consecutive term magnitudes). Rational arithmetic therefore yields

$$\mathrm{Ci}(\pi) > H_5 - \frac{1}{10} - \frac{47}{100} + \sum_{j=1}^{5} \frac{(-1)^j r_j^{2j}}{(2j)(2j)!} > \frac{60}{1000},$$
$$\mathrm{Si}(\pi) > \sum_{j=0}^{3} \frac{(-1)^j r_j^{2j+1}}{(2j+1)(2j+1)!} > \frac{1838}{1000}.$$

Thus the first drift exceeds $(p\cdot 0.060 + 1.838)/2 > 1.01$. □

The discrepancy concerns an exact coefficient in the published conjunction. The following subsection establishes the positive assertions about the threshold and the monotonicity of the first zero. Implicit differentiation and differentiation of Mittag–Leffler series are established tools. The result here identifies their consequence for the stated first-zero law.

6.1 A certified threshold, strict monotonicity and the square-root branch

Write $F(a,t) = E_{a,2}(-t^a)$ and $Y_a(t) = tF(a,t)$. In the journal version of Chaudhary–Diethelm–Hashemishahraki, Conjecture 2(a) asks for positivity below a threshold a_*, and parts (b)(ii)–(iii) ask for a square-root law above the threshold and strict decrease of the first zero. Their cited phase diagram of Hanneken, Achar and Vaught (2013, doi:10.1155/2013/421685, Table 2) already lists the numerical threshold 1.59911520632302. We give rigorous enclosures, prove the global assertions, and determine the square-root coefficient.

Theorem 6.2. *There are a_*, t_* satisfying*

$$|a_* - 1.599115206323013| < 10^{-13}, \qquad |t_* - 5.210687979825734| < 10^{-11}$$

such that $F(a,t) > 0$ for every $1 < a < a_*$ and $t \ge 0$. At a_* there is exactly one positive zero t_*, of multiplicity two. For $a_* < a \le 2$ a first positive zero $Z_2(a)$ exists, and the function Z_2 is strictly decreasing on $[a_*, 2]$. As $a \downarrow a_*$,

$$Z_2(a) = t_* - \sqrt{\frac{-2F_a(a_*,t_*)}{F_{tt}(a_*,t_*)}}\sqrt{a-a_*} + O(a-a_*).$$

The magnitude of this coefficient is between 4.787 and 4.789.

Certified compact estimates. Let A and T be the closed intervals in the first display. The verifier establishes the following facts by outward-rounded arithmetic.

1. The rectangle $A \times T$ contains a simultaneous zero (a_*, t_*) of (F, F_t), and the displayed bounds on the square-root coefficient hold throughout that rectangle.

2. For every $a \in A$, one has $F > 0$ when $t \in [3, 5.20] \cup [5.22, 10]$, and $F_{tt} > 0$ when $5.20 \le t \le 5.22$.

3. For every $a \in [1.5991, 2]$, one has $F > 0$ when $0 \le t \le 3$ and $F_a < 0$ when $3 \le t \le 5.22$.

The finite procedure uses $X = t^a$ and the series $F = \sum_{k\ge 0}(-X)^k/\Gamma(ak+2)$. Differentiate term by term to obtain F_t, F_{tt}, F_a, F_{at}. For $t \ge 1$ their multipliers are

$$\frac{ak}{t}, \quad \frac{ak(ak-1)}{t^2}, \quad k(\log t - \psi(ak+2)), \quad \frac{k + ak^2(\log t - \psi(ak+2))}{t}.$$

For $1.59 \le a \le 2$ and $1 \le t \le 10$, each term with $k \ge 71$ including its multiplier is bounded in absolute value by $20(k+2)^3 10^{1.59k}/\Gamma(1.59k+2)$. Indeed, $10^{ak}/\Gamma(ak+2)$ decreases in a on this range since $\psi(1.59 \cdot 71 + 2) > \log 10$ and ψ is increasing. One may use $\psi(ak+2) \le \log(ak+2)$ to bound the multipliers. The successive majorants have ratio below $1/2$. This follows from $\Gamma(x+s)/\Gamma(x) \ge x^s$ for $x > 0, s \ge 1$, by Jensen's inequality applied to a gamma random variable. Twice the first majorant is less than 10^{-36}. The same bound holds for F on $0 \le t < 1$ and for its X derivative there, since for $k \ge 71$ the maximum of $kX^{k-1}/\Gamma(ak+2)$ occurs at the right edge of the full X interval, with $X \ge 1$. This bound is added to every truncated series result.

For the simultaneous zero, take the midpoint x_0 of $B = A \times T$, write $P = (F, F_t)$ and let M be the interval enclosure for $P'(B)$. Take a fixed dyadic matrix U from an inverse midpoint-Jacobian calculation. The script checks $\det U \ne 0$ and, in each coordinate,

$$|(UP(x_0))_i| + \sum_j |(I - UM)_{ij}| \operatorname{rad}(B_j) < \operatorname{rad}(B_i).$$

The continuous map $x \mapsto x - UP(x)$ therefore maps B into its interior. Brouwer's theorem yields the simultaneous zero there. For the second item, the code checks F using the mean value theorem in t and F_{tt} by direct interval evaluation. It uses 49, 166 and 1 final time intervals on the three ranges, respectively. For the third item, first check $F(a, 3) > 0$ and $F_X < 0$ for $0 \le X \le 3^a$ using 25 final parameter intervals and subdivisions in X. For $F_a < 0$ on the indicated rectangular strip it uses 129 final parameter intervals with subdivisions in t. The dyadic subdivision and all precision and stopping rules are in the source.

Unbounded tail at the critical order. For $1 < a < 2$ Laplace inversion gives

$$Y_a(t) = \frac{2}{a} e^{t\cos(\pi/a)} \cos\bigl(t\sin(\pi/a) - \pi/a\bigr) + \frac{-\sin(\pi a)}{\pi} \int_0^\infty \frac{e^{-rt} r^{a-2}}{1 + 2r^a \cos(\pi a) + r^{2a}}\, dr.$$

Indeed, its transform is $p^{a-2}/(1+p^a)$. Move the contour to the negative real cut, using the principal power. At the two poles $p = e^{\pm i\pi/a}$ the residues of the transform are $1/(ap)$. The jump across the cut gives the displayed positive integral. The large arc vanishes since the transform is $O(|p|^{-2})$, and the arc around zero vanishes since $a > 1$.

For $t \ge 10$ keep only $0 < r < 1/t$ in that integral. It is bounded below, including its coefficient, by $D_a t^{1-a}$ where

$$D_a = \frac{-\sin(\pi a)}{\pi(a-1)e(1+1/10)^2}.$$

For all $a \in A$ the verifier checks $D_a 10^{1-a} > (2/a)e^{10\cos(\pi/a)}$ and $\cos(\pi/a) + (a-1)/10 < 0$. Thus the exponential divided by the power-law lower bound is decreasing on $[10, \infty)$ and already below one at 10. This proves strict positivity on the infinite tail.

Propagation between orders. The compact bounds and the tail imply $Y_{a_*}(t) \ge 0$ everywhere. Indeed, on $[5.20, 5.22]$ the unique critical zero is a strict minimum by strict convexity,

and strict positivity holds off that interval. Let $1 < a < a_*$ and $q = a/a_* < 1$. For the Laplace transforms G_a of Y_a one has

$$G_a(p) = p^{-2(1-q)} G_{a_*}(p^q).$$

Since Y_{a_*} is nonnegative, G_{a_*} is completely monotone. Composition with the Bernstein function p^q shows that $G_{a_*}(p^q)$ is the Laplace transform of a positive measure μ. It satisfies $G_{a_*}(p^q) \sim p^{-2q}$ as $p \to \infty$, so $\mu([0,u]) > 0$ for every $u > 0$. Otherwise its transform would be $O(e^{-(p-1)u})$ by comparison with its value at 1. The inverse transform of $p^{-2(1-q)}$ is positive on all positive times. Its convolution with μ is bounded below by a positive number locally about any $t_0 > 0$: restrict the integral to $[0, t_0/2]$ and use times in $[3t_0/4, 5t_0/4]$. Uniqueness of Laplace transforms and continuity of Y_a give pointwise strict positivity.

At each $a > a_*$ the order-derivative bound gives $F(a, t_*) < 0$. Its first zero therefore belongs to $(3, t_*)$. If $a_* \le a_1 < a_2 \le 2$, evaluate $F(a_2, t)$ at the first zero for a_1. The strict order-derivative bound again gives a negative value, proving $Z_2(a_2) < Z_2(a_1)$. In particular the certified touching parameter is the unique transition order.

As $a \downarrow a_*$ compactness and strict positivity before t_* force $Z_2(a) \to t_*$. At the touching point, $F_t = 0$, $F_{tt} > 0$ and $F_a < 0$. Put $a = a_* + s^2$ and $t = t_* + sv$. After division by s^2 the local equation extends analytically to $s = 0$ with value $F_a + F_{tt} v^2/2$. Both its roots in v are simple. The implicit function theorem and local strict convexity identify the branch with the negative square root as the first zero and prove the displayed expansion.

7 The Guo–Qi conjecture for a normalized gamma-ratio derivative

B.-N. Guo and F. Qi, *A class of completely monotonic functions involving divided differences of the psi and tri-gamma functions and some applications*, J. Korean Math. Soc. **48** (2011), 655–667, doi:10.4134/JKMS.2011.48.3.655, conjecture in Remark 4 the logarithmic complete monotonicity of their function $\Phi_{s,t}$ from (36) for $|s-t| > 1$ and of its reciprocal for $|s-t| < 1$. This includes the diagonal $s = t$. The 2009 preprint (arXiv:0903.1430) states the conjecture in Remark 3.

Write x for the original argument translated by $\min(s,t)$, so its entire domain is $x > 0$, and put $r = |s-t|$. The original function in these coordinates is $\Phi_r = P_r'$, where

$$P_r(x) = \begin{cases} [\Gamma(x+r)/\Gamma(x)]^{1/r}, & r > 0, \\ \exp \psi(x), & r = 0. \end{cases}$$

In particular $\Phi_0(x) = \exp(\psi(x))\psi'(x)$. We use the standard sign convention that a positive function Φ is logarithmically completely monotone precisely when $-(\log \Phi)'$ is completely monotone.

Lemma 7.1 (Comparing a Laplace density with its canonical measure)**.** *Let k be a strictly positive C^2 function on $[0,\infty)$ with $k(0) = 1$. Suppose $F(x) = \int_0^\infty e^{-xv} k(v)\,dv$ is finite for every $x > 0$, and $H = -F'/F$ is completely monotone. Its positive representing measure has a continuous density $h \ge 0$. With $\ell = (\log k)'$, the exact identity*

$$h(v) - k(v) = \int_0^v [\ell(v) - \ell(v-u)] k(v-u) h(u)\,du$$

holds for $v \ge 0$. Consequently $H - F$ is completely monotone for log-convex k, and $F - H$ is completely monotone for log-concave k.

Proof. By the Bernstein–Widder theorem let H be the Laplace transform of the positive measure μ. The product $FH = -F'$ and uniqueness of Laplace transforms give

$$(k * \mu)(v) := \int_{[0,v]} k(v-u)\, d\mu(u) = vk(v)$$

for almost every $v > 0$. Both sides are right-continuous, so equality holds for every $v \geq 0$, with $\mu(\{0\}) = 0$. Direct integration of $k(w) = 1 + \int_0^w k'(s)\, ds$ gives

$$(k * \mu)(v) = \mu([0,v]) + \int_0^v (k' * \mu)(s)\, ds.$$

All integrations on compact intervals are valid by local boundedness. It follows that μ is absolutely continuous and its density satisfies $h(v) = k(v) + vk'(v) - (k' * h)(v)$ almost everywhere. The right hand side is continuous, so it gives a continuous nonnegative version of h, with $h(0) = 1$. Inserting $vk'(v) = \ell(v)(k * h)(v)$ yields the identity and its two sign consequences. Laplace transformation is legitimate since both h and k have finite transforms at every $x > 0$. □

Theorem 7.2. *For every $r > 1$, Φ_r is logarithmically completely monotone on $(0, \infty)$. For every $0 \leq r < 1$, $1/\Phi_r$ is logarithmically completely monotone there. All the logarithmic derivative inequalities are strict in these two ranges. At $r = 1$, $\Phi_1 = 1$.*

Proof. Put $F_r = (\log P_r)'$. For $r > 0$ the trigamma series yields

$$F_r(x) = \frac{\psi(x+r) - \psi(x)}{r} = \frac{1}{r} \int_0^r \sum_{n \geq 0} \frac{du}{(x+n+u)^2}, \qquad F_0(x) = \sum_{n \geq 0} \frac{1}{(x+n)^2}.$$

These positive integral and sum representations put every F_r in the generalized Stieltjes class of order two. We invoke G. K. Kristiansen, *A proof of Steutel's conjecture*, Ann. Probab. **22** (1994), 442–452, in its standard consequence that every nonzero generalized Stieltjes function of order two is logarithmically completely monotone. For the explicit class formulation see C. Berg, S. Koumandos and H. L. Pedersen, *Nielsen's beta function and some infinitely divisible distributions*, Theorem 2.1, arXiv:1905.04131. In particular $H_r = -F_r'/F_r$ is completely monotone. This input supplies positivity of the canonical measure. No positivity of the difference $H_r - F_r$ follows without a further comparison.

Using $(x+a)^{-2} = \int_0^\infty v e^{-(x+a)v}\, dv$ gives F_r as the Laplace transform of the normalized kernel

$$k_r(v) = \frac{1 - e^{-rv}}{r(1 - e^{-v})} \quad (r > 0), \qquad k_0(v) = \frac{v}{1 - e^{-v}}, \quad k_r(0) = 1.$$

For $v > 0$ and $r > 0$,

$$(\log k_r)''(v) = \frac{1}{4 \sinh^2(v/2)} - \frac{r^2}{4 \sinh^2(rv/2)}.$$

The strict increase of $z \mapsto \sinh(z)/z$ on $(0, \infty)$ shows that this is positive for $r > 1$ and negative for $0 < r < 1$. At $r = 0$ the second term is $1/v^2$, giving the same negative sign. Lemma 7.1 therefore supplies complete monotonicity of $H_r - F_r$ for $r > 1$ and of $F_r - H_r$ for $r < 1$. Because $\Phi_r = P_r F_r$, these are exactly $-(\log \Phi_r)'$ and $-(\log(1/\Phi_r))'$, respectively. Strictness follows from $h(0) = 1$, continuity, and the strict change of ℓ on every positive interval. At $r = 1$ the gamma recurrence gives $P_1(x) = x$. □

Comparison with an earlier incompatible statement. Y.-C. Chen, T. Mansour and Q. Zou, *On the complete monotonicity of quotient of gamma functions*, Math. Inequal. Appl. **15** (2012), 395–402, doi:10.7153/mia-15-33, Theorem 1, asserted failure of ordinary complete monotonicity of $P_{s,t}$, identified in their (5) with the derivative $\Phi_{s,t}$. Their claim for $|s-t|>1$ can be checked directly at $r=2$. There $\Phi_2=(2x+1)/(2\sqrt{x(x+1)})$ and the inverse Laplace density of $-(\log\Phi_2)'$ is $(1+e^{-v})/2-e^{-v/2}=(1-e^{-v/2})^2/2\ge 0$. Hence Φ_2 is logarithmically completely monotone and consequently completely monotone. The preceding theorem establishes the entire parameter range of the 2011 conjecture.

8 The optimal reciprocal gamma-quotient threshold

Qi, Wei and Guo, *Complete monotonicity of a function involving the ratio of gamma functions and applications*, Banach J. Math. Anal. **6** (2012), no. 1, 35–44, doi:10.15352/bjma/1337014663, define, for $y>-1$, $b=y+1$ and $x>-b$,

$$h_{a,y}(x)=(x+b)^{-a}\exp\left(\frac{\log\Gamma(x+b)-\log\Gamma(b)}{x}\right).$$

The quotient in the exponential takes the value $\psi(b)$ at $x=0$. A positive function h is logarithmically completely monotone (LCM) when $(-1)^k(\log h)^{(k)}\ge 0$ for every $k\ge 1$. The authors' Remark 4.5 conjectures that, if $b>1/2$, LCM of $1/h_{a,y}$ on its entire domain requires $a\le 1/(2b)$. This statement also appeared as Remark 5 in the 2009 Qi–Guo preprint, arXiv:0904.1101.

Theorem 8.1. *For every $b>0$, the reciprocal $1/h_{a,b-1}$ is LCM on $(-b,\infty)$ if and only if $a\le A(b)$. The sharp threshold equals 1 when $b\le 1/2$. When $b>1/2$, define*

$$B_b(s)=\frac{s}{(1+bs)(1-e^{-s})}.$$

There is a unique positive root t_b of

$$\int_{t_b}^{\infty}B_b'(s)\,\frac{e^{-bs}}{s}\,ds=0.$$

Then $A(b)=B_b(t_b)$, and

$$\frac{2}{1+2b}<A(b)<\min\{1,1/b\}.$$

In particular, the conjectured bound fails for every $b>1/2$. One explicit counterexample has $y=0$ and $a=2/3$.

Proof. Translate the full domain by putting $X=x+b>0$. Write

$$F_b(X)=\frac{\log\Gamma(X)-\log\Gamma(b)}{X-b}=\int_0^1\psi\big(b(1-u)+uX\big)\,du.$$

This also gives the removable value at $X=b$. The logarithm of the reciprocal is $a\log X-F_b(X)$. Using the trigamma integral in F_b' and then $t=us$ gives

$$F_b'(X)=\int_0^1 u\int_0^\infty\frac{se^{-b(1-u)s-uXs}}{1-e^{-s}}\,ds\,du=\int_0^\infty e^{-Xt}K_b(t)\,dt,$$
$$K_b(t)=te^{bt}\int_t^\infty\frac{e^{-bs}}{s(1-e^{-s})}\,ds.$$

The exchange uses Tonelli's theorem for a nonnegative integrand. Thus the negative logarithmic derivative of the reciprocal is the Laplace transform of $K_b(t)-a$. Bernstein's theorem and

uniqueness of Laplace transforms imply that it is completely monotone exactly when $K_b(t) \geq a$ everywhere. Indeed the signed density is continuous and bounded as shown below; after any fixed positive exponential tilt both it and a positive representing measure are finite. Consequently $A(b) = \inf_{t>0} K_b(t)$.

Set $W_b(t) = e^{-bt}/t$. The last kernel is the normalized tail average

$$K_b(t) = \frac{1}{W_b(t)} \int_t^\infty B_b(s)\,[-W_b'(s)]\,ds.$$

The bounded function B_b has limits 1 and $1/b$ at zero and infinity. The same endpoint limits hold for K_b. At infinity this follows from averaging on $[t, \infty)$. At zero the mass of $[\varepsilon, \infty)$ in the normalized average is $W_b(\varepsilon)/W_b(t) \to 0$ for every fixed $\varepsilon > 0$.

Logarithmic differentiation shows

$$\frac{B_b'(s)}{B_b(s)} = \frac{1}{s(1+bs)} - \frac{1}{e^s - 1}.$$

The power series of $(e^s - 1 - s)/s^2$ has positive coefficients, starts at $1/2$, and increases strictly to infinity. Thus B_b is strictly increasing for $b \leq 1/2$, in which case its tail average is also increasing with infimum 1. For $b > 1/2$, B_b decreases until a unique point s_b and then increases. Moreover $B_b'(0+) = 1/2 - b < 0$. Integration by parts gives

$$K_b(t) - B_b(t) = \frac{1}{W_b(t)} \int_t^\infty B_b'(s) W_b(s)\,ds, \qquad K_b'(t) = \frac{-W_b'(t)}{W_b(t)}(K_b(t) - B_b(t)).$$

The integral is positive for $t \geq s_b$. On $(0, s_b)$ it increases strictly from $-\infty$ to a positive number. Its negative divergence follows from $W_b(s) \sim 1/s$ and $B_b'(0+) < 0$. Hence exactly one root t_b exists. The kernel decreases strictly up to t_b and increases strictly thereafter, proving the characterization and the upper bound for its minimum.

Finally $1/(1-e^{-s}) > 1$ and $1/(1-e^{-s}) = \frac{1}{2} + \frac{1}{2}\coth(s/2) > 1/s + 1/2$ for $s > 0$. Consequently

$$B_b(s) > \frac{\max\{s, 1 + s/2\}}{1 + bs}.$$

For $b > 1/2$ the right side decreases on $(0, 2)$ and increases on $(2, \infty)$, so its minimum is $2/(1 + 2b)$. Applying this strict bound at t_b finishes the proof, since $2/(1 + 2b) > 1/(2b)$. □

The auxiliary problem was to locate the minimum of a normalized tail average with a one-valley integrand. Its derivative is determined by a tail average of the derivative of that integrand. This observation gives a unique critical point and a sharp implicit threshold. All ingredients are elementary or classical. Historical priority for the characterization is a separate question from the refutation.

9 The two squared-tangent remainder questions

Write $a_j = [x^{2j}] \tan^2 x$ for $j \geq 1$, and set

$$U_n(x) = \frac{\tan^2 x - \sum_{j=1}^{n-1} a_j x^{2j}}{a_n x^{2n}}, \qquad U_n(0) = 1.$$

This is $T_{2n-1}[\tan^2 x]$ in the notation of F. Qi, *Theory of normalized remainders in Taylor series expansions*, arXiv:2512.15797v4, Section 12.7.1. Its First Conjecture (X.-L. Liu and F. Qi, Math. Slovaca 75 (2025), Remark 4.1) asks whether U_{n+1}/U_n is concave on $(-\pi/2, \pi/2)$. The Second Conjecture (Y. Shuang, C.-Y. He and F. Qi, J. Math. Inequal. 19 (2025), Section 4.1) asks whether U_n/U_{n+1} is absolutely monotone on $(0, \pi/2)$ and completely monotone on $(-\pi/2, 0)$. They are separately posed questions about opposite quotients. We give one proof for both.

Theorem 9.1. *For every $n \geq 1$, the Maclaurin coefficients of U_{n+1}/U_n at every strictly positive even degree are negative. Those of U_n/U_{n+1} at every such degree are positive. Both expansions converge for $|x| < \pi/2$. Consequently both conjectures hold, with strict concavity in the first and strict derivative signs in the second.*

A coefficient criterion. Suppose $D(y) = \sum_{j \geq 0} d_j y^j$ is real analytic in a disk of radius greater than $\rho > 1$, with $d_0 > 0$, $D(\rho) = 0$, and $d_j \geq 0$ for every $j \geq 2$. Define

$$r = 1/\rho, \qquad e = \frac{1}{d_0} \sum_{j \geq 2} (j-1) d_j \rho^j.$$

If $0 \leq e < 1$ and $2re/(1-e) < (1-r)^2$, then all coefficients of degrees $k \geq 2$ in $(1-y)^2/D(y)$ are positive and this series is analytic on $|y| < \rho$.

Indeed, put $Q(y) = D(y)/(1-ry)$. Its constant is d_0, and for $k \geq 1$ its coefficient is

$$[y^k]Q(y) = -\sum_{j>k} d_j \rho^{j-k} \leq 0, \qquad Q(\rho) = -\rho D'(\rho) = d_0(1-e) > 0.$$

Thus the geometric expansion of $1/Q$ has nonnegative coefficients b_j, and converges on $|y| \leq \rho$, because the sum of the absolute nonconstant coefficients of Q/d_0 weighted by ρ^j equals $e < 1$. In particular

$$\sum_{j \geq 1} b_j \rho^j = b_0 e/(1-e).$$

If $v_j = [y^j](1/D)$, then $v_j = r v_{j-1} + b_j$ and $v_j \geq b_0 r^j$. For $k \geq 2$,

$$v_k - 2v_{k-1} + v_{k-2} = (1-r)^2 v_{k-2} - (2-r) b_{k-1} + b_k \geq b_0 r^{k-2}\left[(1-r)^2 - 2re/(1-e)\right] > 0.$$

This proves the criterion.

Proof of Theorem 9.1. Put $y = (2x/\pi)^2$ and $c = n + 1/2$. The partial fractions for $\sec^2 x$, or equivalently the derivative of the tangent partial fractions, give

$$U_n(x) = F_n(y)/F_n(0), \qquad F_n(y) = \sum_{j \geq 0} (c+j) y^j \sum_{m \geq 1,\ m \text{ odd}} m^{-2n-2-2j}.$$

The constant term of $\sec^2 x$ does not enter since $n \geq 1$. For $n \geq 2$, separate the $m = 1$ pole and write

$$D(y) = (1-y)^2 F_n(y) = c - (c-1)y + (1-y)^2 H(y),$$
$$H(y) = \sum_{m \geq 3,\ m \text{ odd}} m^{-2n-2} \left(\frac{c}{1-uy} + \frac{uy}{(1-uy)^2} \right), \qquad u = m^{-2},$$

where u inside the sum depends on m. The secondary term H is analytic on $|y| < 9$. Its total weight $s = \sum_{m \geq 3,\ m \text{ odd}} m^{-2n-2}$ satisfies

$$s \leq 3^{-2n-2} \left(1 + \frac{3}{2(2n+1)} \right) < 2 \cdot 9^{-(n+1)}$$

by integral comparison with mesh two. As $u \leq 1/9$ and $c \geq 5/2$,

$$H(0) = cs,$$
$$H(2) \leq (9c/7 + 18/49)s < 3cs/2,$$
$$H'(2) \leq (9(c+1)/49 + 36/343)s < cs/3.$$

The coefficients d_j of D are positive for $j \geq 2$. In fact each u term contributes its weight times $u^{j-2}(1-u)((c+j-2)(1-u) - 2u) > 0$. Now $D(1) = 1$ and $D(2) = 2 - c + H(2) < 0$, so

there exists $1 < \rho < 2$ with $D(\rho) = 0$. The equation at this zero implies $c(\rho - 1) \geq \rho$, whence $1 - 1/\rho \geq 1/c$. Moreover

$$\begin{aligned} d_0 e &= \rho D'(\rho) - D(\rho) + d_0 \\ &= H(0) + (\rho^2 - 1)H(\rho) + \rho(\rho - 1)^2 H'(\rho) < 7cs. \end{aligned}$$

Since $d_0 \geq c$, we have $e < 14 \cdot 9^{-(n+1)} < 1/(2c^2 + 1)$. For the last inequality check $n = 2$ and note that $2(c + 1)^2 + 1 < 9(2c^2 + 1)$. Thus $2re/(1 - e) < 1/c^2 \leq (1 - r)^2$. The criterion proves $[y^k](1/F_n) > 0$ for $k \geq 2$.

For $n = 1$, $1/U_1(x) = x^2 \cot^2 x = x^2 \csc^2 x - x^2$. The usual partial fractions $\csc^2 x = \sum_{j \in \mathbb{Z}} (x - j\pi)^{-2}$ give strictly positive coefficients in $x^2 \csc^2 x$ at every positive even degree. Consequently the same strict positivity holds in $1/F_1$ from degree two in y onward.

Finally $F_n(y) = F_n(0) + yF_{n+1}(y)$, so

$$R(y) := \frac{U_{n+1}(x)}{U_n(x)} = \frac{F_n(0)}{F_{n+1}(0)y} \left(1 - \frac{F_n(0)}{F_n(y)}\right).$$

It has constant 1 and all higher coefficients negative. Its expansion converges in $|y| < 1$ by the above representations. For $0 \leq y < 1$ the ratio itself is positive. Therefore $|1 - R(y)| \leq 1 - R(|y|) < 1$ throughout this disk. The expansion $1/R = \sum_{j \geq 0} (1 - R)^j$ then has strictly positive coefficients of every degree. Substituting $y = 4x^2/\pi^2$ and differentiating proves the required strict curvature and derivative signs. Evenness gives the alternating signs on the negative half-interval. □

10 Sharp constants for He–Qi's beta-product majorization problem

C.-Y. He and F. Qi, *Integral representations for specific Maclaurin series and their applications to normalized remainder of exponential function*, Open J. Math. Sci. **10** (2026), 296–305, doi:10.30538/oms2026.0287, pose Problem 3 for

$$S_{m,j}(x) = \sum_{h=0}^{\infty} \binom{h + m + j}{j}^{-1} \frac{x^h}{h!}, \qquad m, j \in \mathbb{N}_0, \quad x \in \mathbb{R}.$$

The beta integral in their Theorem 5 gives $S_{m,j}(x) = j \int_0^1 t^m (1 - t)^{j-1} e^{xt}\, dt$ for $j > 0$, while $S_{m,0}(x) = e^x$. We will also use the same integral to extend the shape j to positive reals.

Theorem 10.1. *The two assertions of absolute monotonicity in He–Qi Problem 3, for its functions G and H in (27)–(28), hold on the full real line. For its remaining two functions in (29)–(30), write $s = (p, q)$, $t = (a, b)$ in decreasing order, where s strictly majorizes t, so $p > a \geq b > q \geq 0$ and $p + q = a + b$. Put $P_s = S_{m,p} S_{m,q}$ and $P_t = S_{m,a} S_{m,b}$. The largest constants for which $P_t - AP_s$ and $P_s - BP_t$, respectively, are absolutely monotone on $\mathbb{R}$ are*

$$A_* = \frac{a!\, b!}{p!\, q!}, \qquad B_* = \frac{pq}{ab}.$$

For $q = 0$, the sharp value of B is zero, whereas the source asks for $0 < B < 1$. No positive B has the stated property in this case. For $q > 0$ both optimal constants lie strictly between zero and one. The formulas hold for every integer $m \geq 0$.

Proof. Any transform $\int_{[0,2]} e^{xv} d\sigma(v)$ of a finite positive measure is absolutely monotone on $\mathbb{R}$, by differentiation under the integral. We first address

$$G_{m,n;k,l} = S_{m,n+l} S_{m+k,n} - S_{m+k,n+l} S_{m,n}, \quad k, l \geq 1.$$

For $n > 0$, expand the two beta integrals and interchange the two integration variables in half of the difference. The resulting integrand, apart from a positive constant and the common factor $e^{x(t+u)}(tu)^m[(1-t)(1-u)]^{n-1}$, is

$$(u^k - t^k)\{(1-t)^l - (1-u)^l\} \geq 0.$$

For $n = 0$ the difference equals $e^x(S_{m,l} - S_{m+k,l})$, whose integrand contains the nonnegative factor $t^m(1-t^k)$.

The other assertion in (28) concerns $H = S_{p,n+1}S_{q,n+1} - S_{a,n+1}S_{b,n+1}$. After symmetrization its sign is given pointwise by $t^p u^q + t^q u^p - t^a u^b - t^b u^a \geq 0$. Indeed, for $t, u > 0$ the two symmetrized sums are $2(tu)^{(p+q)/2}\cosh(d\log(t/u)/2)$ for $d = p-q$ and $a-b$, respectively. Since $p - q > a - b$, the inequality follows.

For the sharp constants, suppose first that $q > 0$. In each product integral substitute $v = 1-t$, $w = 1-u$ and group by $y = v + w$, so $0 < y < 2$ and the exponential is $e^{x(2-y)}$. Put $z = v/y$. For shapes (i, j) the density on this slice is

$$ij\, y^{i+j-1} \int_0^1 W_y(z) z^{i-1}(1-z)^{j-1}\, dz, \qquad W_y(z) = \mathbf{1}_{\{yz\leq 1,\ y(1-z)\leq 1\}}[(1-yz)(1-y(1-z))]^m.$$

We set $W_y = 0$ outside the indicated set. On that set the base of the power is $1 - y + y^2 z(1-z)$. Consequently W_y is symmetric and nonincreasing with $|z - 1/2|$, also when $m = 0$.

Use the symmetric probability measure proportional to $z^{a-1}(1-z)^{b-1} + z^{b-1}(1-z)^{a-1}$ on $(0, 1)$. The ratio of the corresponding symmetrized s and t integrands is

$$R(z) = \frac{\cosh((p-q)c)}{\cosh((a-b)c)}, \qquad c = \tfrac{1}{2}|\log(z/(1-z))|.$$

It is at least one and increases with $|z - 1/2|$, since $d\tanh(dc)$ is nondecreasing in $d \geq 0$. Both the unweighted integral of R and the weighted one are finite because $q > 0$. Opposite monotonicity gives $\mathbb{E}(W_y R) \leq \mathbb{E}(W_y)\mathbb{E}(R)$, by integrating $(W_y(z) - W_y(z'))(R(z) - R(z')) \leq 0$ against two independent copies. Here $\mathbb{E}(W_y) > 0$. Therefore the ratio of the P_s to P_t slice densities lies between

$$\frac{pq}{ab} \quad \text{and} \quad \frac{pq}{ab}\frac{\mathrm{B}(p,q)}{\mathrm{B}(a,b)} = \frac{\Gamma(p+1)\Gamma(q+1)}{\Gamma(a+1)\Gamma(b+1)}.$$

Both claimed differences have nonnegative densities on every slice. This proves their absolute monotonicity, with the displayed constants, also for positive real shapes using gamma factors.

As $j \downarrow 0$ the measures $jv^{j-1}\, dv$ on $[0, 1]$ converge weakly to the atom at zero. Hence the continued $S_{m,j}$ and every one of its derivatives converge to e^x for fixed x. In the preceding argument take the spread shapes $(p - \varepsilon, \varepsilon)$ and let $\varepsilon \downarrow 0$, keeping their sum fixed. This gives the asserted nonnegative differences at $q = 0$, with $0! = 1$ and $B_* = 0$.

It remains to prove optimality. For each $j > 0$ the beta integral gives

$$S_{m,j}(x) \sim e^x \Gamma(j+1) x^{-j} \quad (x \to +\infty), \qquad S_{m,j}(-X) \sim j\,\Gamma(m+1) X^{-m-1} \quad (X \to +\infty).$$

For the first estimate substitute $x(1-t)$ as variable and use dominated convergence. For the second substitute Xt on $0 < t < 1/2$, where $(1-t)^{j-1}$ is bounded, and bound the remaining beta integral by an exponentially small tail. The first estimate continues to hold for $j = 0$ by the exact formula $S_{m,0} = e^x$. Thus $P_t/P_s \to A_*$ as $x \to +\infty$ and $P_s/P_t \to B_*$ as $x \to -\infty$, the second ratio tending to zero when $q = 0$. Larger constants violate positivity of the undifferentiated function. For $q > 0$, strict concavity of log gives $pq < ab$, and strict convexity of $\log\Gamma(j+1)$ gives $\Gamma(a+1)\Gamma(b+1) < \Gamma(p+1)\Gamma(q+1)$. This proves all statements of the theorem. □

Corollary 10.2 (Finite products with real shapes)**.** *For real $m \geq 0$, let s, t be positive real vectors of the same finite length, with s majorizing t in the usual weak-partial-sum sense and*

with equal totals. For $P_v = \prod_i S_{m,v_i}$ the largest constants such that $P_t - AP_s$ and $P_s - BP_t$ are absolutely monotone on $\mathbb{R}$ are

$$A = \prod_i \frac{\Gamma(t_i+1)}{\Gamma(s_i+1)}, \qquad B = \prod_i \frac{s_i}{t_i}.$$

Proof. The Hardy–Littlewood–Pólya two-coordinate characterization of majorization (see A. W. Marshall, I. Olkin and B. C. Arnold, *Inequalities: Theory of Majorization and Its Applications*, 2nd ed., Springer, 2011, Chapter 2, Sect. B) gives a finite chain from s to t by permutations and replacements $(p,q) \mapsto (\lambda p + (1-\lambda)q, (1-\lambda)p + \lambda q)$, where $0 \le \lambda \le 1$. Every intermediate shape is positive. The slice proof applies to real $m \ge 0$ and real positive shapes. Multiply its two-coordinate inequalities in the order of absolute monotonicity by the untouched factors, which are themselves absolutely monotone by the beta integral. Compose the resulting inequalities along the chain. This is valid because the cone of absolutely monotone functions is closed under addition and multiplication. The constants telescope to the displayed products. The endpoint asymptotics in the proof above show sharpness for the entire product, since the sums of all shapes agree. □

Qi's Fifth Conjecture. The "Fifth Conjecture" in F. Qi, arXiv:2512.15797, Section 12.7.1.5, asks for absolute monotonicity of the three special products (12.7.1)–(12.7.3). Its G_n is $G_{0,n;1,1}$, its H_n is the variance minor of the single beta weight with parameter n, and its J_n is the sharp A_* inequality with $m = 1$, $(p,q) = (n+2, n)$, $(a,b) = (n+1, n+1)$. The above theorem therefore settles the three clauses together, including the endpoint $n = 0$ for J_n. Their original statements appear in Qi, Mathematics **12** (2024), 2859, Remark 7, doi:10.3390/math12182859, and in the cited paper of He and Qi. These specializations are counted together with the general problem as one contribution.

For a direct view of their dependence, put

$$D_p(x) = \int_0^1 \int_0^1 (t-u)^2 [(1-t)(1-u)]^p e^{x(t+u)}\, dt\, du \qquad (p \ge 0).$$

With the chapter's indexing, its three left sides satisfy

$$G_n = \tfrac{n(n+1)}{2} D_{n-1}, \qquad H_n = \tfrac{n^2}{2} D_{n-1} \quad (n \ge 1), \qquad J_n = \tfrac{n+1}{2} D_n \quad (n \ge 0).$$

The first two follow by symmetrizing the beta integrals. For the third, take $n > 0$ and put $v = 1-t$, $w = 1-u$. After symmetrization on each fixed $y = v + w$ the integrand, besides the factor $(n+1)e^{x(2-y)}$, is

$$(1-v)(1-w)(vw)^{n-1}\big[vw - \tfrac{n}{2}(v-w)^2\big] = \tfrac{1}{2}(1-v)(1-w)\frac{d}{dv}\big[(v-w)(vw)^n\big].$$

Here $w = y - v$. At a slice endpoint one of $v, w, 1-v, 1-w$ vanishes. Integration by parts therefore gives $\frac{1}{2}\int (v-w)^2 (vw)^n\, dv$ on that slice. The limit $n \downarrow 0$ follows from the same weak convergence of beta measures used above. These identities also prove strict positivity of every derivative of the three special differences.

Earlier determinant formulas. Baricz and Ismail, *Turán type inequalities for Tricomi confluent hypergeometric functions*, Remark 9, arXiv:1110.4699, display squared-Vandermonde integral formulas for Kummer determinants, including the kernel D_p above, and credit earlier results of Ismail and Laforgia. Although the cited monotonicity statement is phrased for $x > 0$, their displayed positive integral extends directly to real x. The determinant positivity itself is therefore classical. The main comparison here is between two majorized shape vectors on an entire constant-sum slice, with sharp bounds in both directions and products of arbitrary finite length.

11 A large-power obstruction for normalized arcsine remainders

Feng Qi's *Theory of normalized remainders in Taylor series expansions*, arXiv:2512.15797, Section 12.7.1.6 (Sixth Conjecture), asks whether $T_{2n+1}[(\arcsin x/x)^q]$ is logarithmically absolutely monotone on $[0,1)$ for every positive integer q and every integer $n \geq 0$. The author attributes the question to Section 1 of *Series and connections among central factorial numbers, Stirling numbers, inverse of Vandermonde matrix, and normalized remainders of Maclaurin series expansions*, Mathematics **13** (2025), 223, doi:10.3390/math13020223. The required signs are nonnegativity of all positive-order derivatives of the logarithm. We give an exact counterexample and explain how the truncation index and the exponent play different roles.

Put $y = x^2$, $B(y) = \arcsin(\sqrt{y})/\sqrt{y}$ with $B(0) = 1$, and write $B(y)^q = \sum_{k\geq 0} u_k(q) y^k$. For $s = n+1$ the original normalized remainder is exactly $R_{q,s}(x^2)$, where

$$R_{q,s}(y) = \sum_{j\geq 0} \frac{u_{s+j}(q)}{u_s(q)} y^j.$$

Every denominator here is positive since B has positive Taylor coefficients. If $[y^j] \log R_{q,s}(y) < 0$, then the derivative of order $2j$ of its logarithm in x is negative at zero and also for all sufficiently small positive x.

Theorem 11.1. *The Sixth Conjecture is false. At $n = 0$ and $q = 56$ one has*

$$\frac{d^8}{dx^8} \log T_1[(\arcsin x/x)^{56}]\bigg|_{x=0} = -\frac{50349478}{86625}.$$

For every fixed $n \geq 0$ the conjectured property fails for all sufficiently large integers q. Conversely, at each fixed positive integer q it holds for every sufficiently large n.

The finite counterexample. For $s = 1$, $R_{q,1}(y) = (B(y)^q - 1)/(qy/6)$. If $a_k = \binom{2k}{k}/[4^k(2k+1)]$ are the coefficients of B, the identity $B(B^q)' = qB'B^q$ gives $u_0 = 1$ and

$$ku_k = \sum_{j=1}^{k} ((q+1)j - k) a_j u_{k-j}.$$

Put $r_k = u_{k+1}/u_1$ and $l_k = [y^k] \log R_{q,1}(y)$. Logarithmic differentiation gives $kl_k = kr_k - \sum_{j=1}^{k-1} j l_j r_{k-j}$. Substituting $k \leq 4$ from these two finite recurrences gives

$$l_4 = -\frac{q^4}{3732480} + \frac{10181q^2}{16329600} + \frac{2497q}{226800} + \frac{765187349}{15717240000}.$$

At $q = 56$ this equals $-25174739/1746360000$, which proves the stated derivative. The polynomial has just one positive root by Descartes' rule of signs, and it is positive at 55. Thus 56 is the first integer exponent detected by this particular coefficient.

Transfer from a bounded moment distribution. The following argument works for any real analytic base $B(y)$ at zero with $B(0) = 1$ and $B'(0) = a > 0$. For fixed k, binomial expansion of $(1 + (B-1))^q$ gives $u_k(q) = (aq)^k/k! + O(q^{k-1})$. The term of degree k in q comes from $(B-1)^k$. For each fixed $s \geq 1$ the coefficientwise limit as $q \to \infty$ is therefore

$$R_{q,s}(t/(aq)) \longrightarrow M_s(t) = \sum_{j\geq 0} \frac{s!}{(s+j)!} t^j = \mathbb{E}e^{tU},$$

where U has density $s(1-u)^{s-1}$ on $(0,1)$. The beta integral verifies the last equality. The logarithms converge coefficientwise too, since each coefficient depends polynomially on finitely many coefficients of a series with constant term one.

For every s, at least one coefficient of $\log M_s$ is negative. Indeed write $\log M_s(t) = \sum_{j\geq 1} \kappa_j t^j/j!$. Here $\kappa_2 = v = s/[(s+1)^2(s+2)] > 0$. If $\kappa_1, \ldots, \kappa_{2N}$ were all nonnegative, exponentiation would give

$$\mathbb{E}U^{2N} \geq \frac{(2N)!}{2^N N!}\, v^N \geq N! v^N \geq (Nv/e)^N.$$

Taking $N = \lceil 3/v \rceil$ contradicts $U \leq 1$. A strictly negative coefficient consequently persists in the scaled tails at every sufficiently large q. Positive scaling preserves its sign. For $s = 1$ this mechanism has a particularly short witness: $M_1(t) = (e^t - 1)/t$ and $[t^4] \log M_1 = -1/2880$.

The opposite limit with the exponent fixed. The elementary identity

$$B(y) = \frac{\pi/2 - \arcsin\sqrt{1-y}}{\sqrt{y}}$$

holds near $y = 1$ in its slit domain. The representation $B(y) = \int_0^1 (1 - yu^2)^{-1/2}\, du$ gives analytic continuation on $\mathbb{C} \setminus [1, \infty)$. For each fixed q the convergent local expansion in powers of $\sqrt{1-y}$ has first nonconstant half-integer coefficient $-q(\pi/2)^{q-1}$ at exponent $1/2$. Standard coefficient transfer for a slit-domain expansion (or singularity analysis term by term through exponent $7/2$) gives

$$u_k(q) = \frac{q(\pi/2)^{q-1}}{2\sqrt{\pi}}\, k^{-3/2} \left(1 + \frac{b_q}{k} + \frac{c_q}{k^2} + O_q(k^{-3})\right).$$

Consequently the centered second difference of $\log u_k(q)$ is $3/(2k^2) + O_q(k^{-3}) > 0$ for large k.

We recall the short reciprocal-series argument applied to any sufficiently late tail. Its coefficients $v_j = u_{s+j}/u_s$ are positive, $v_0 = 1$, and v_j/v_{j-1} is nondecreasing. Set $1 - 1/R_{q,s}(y) = \sum_{j\geq 1} p_j y^j$. Then $p_1 = v_1 > 0$ and $p_j = v_j - \sum_{i=1}^{j-1} p_i v_{j-i}$. If $p_i \geq 0$ for $i < j$, bound $v_{j-i} \leq (v_j/v_{j-1}) v_{j-1-i}$ in this sum and use the same convolution at $j-1$ to infer $p_j \geq 0$. This is the classical Kaluza sign argument. It also gives $0 \leq p_j \leq v_j$. Therefore for $0 \leq y < 1$ the series $P(y) = \sum p_j y^j = 1 - 1/R_{q,s}(y)$ takes values in $[0,1)$. Expanding $\log R_{q,s} = \sum_{k\geq 1} P^k/k$ now proves nonnegativity of all Taylor coefficients, with convergence throughout $|y| < 1$. Substituting $y = x^2$ proves logarithmic absolute monotonicity through the entire original interval.

12 Log-concavity of the normalized arctangent remainder

The Seventh Conjecture in Section 12.7.1.7 of Feng Qi, *Theory of normalized remainders in Taylor series expansions* (arXiv:2512.15797), following X.-L. Liu and F. Qi, *Logarithmically absolute monotonicity and series expansions of normalized remainders for inverse sine and tangent functions*, asks whether $T_{2n}[\arctan x]$ is logarithmically concave on $(0,1)$ for every $n \geq 0$. Finite geometric division of $(1+t^2)^{-1}$ and integration from zero to x give

$$T_{2n}[\arctan x] = (2n+1) \int_0^1 \frac{t^{2n}}{1 + x^2 t^2}\, dt.$$

Theorem 12.1. *For every real $a > 0$, the function*

$$F_a(x) = a \int_0^1 \frac{t^{a-1}}{1 + x^2 t^2}\, dt$$

satisfies $(\log F_a)''(x) < 0$ for $0 < x \leq 1$. In particular the Seventh Conjecture holds for all its stated indices.

Proof. Put $d = 2x^2/(1+x^2)$ and $r = a/[(1+x^2)F_a(x)]$. Differentiating $x^a F_a(x) = a\int_0^x u^{a-1}/(1+u^2)\,du$ gives $xF_a'/F_a = r - a$ and then $xr' = r(a - d - r)$. Consequently

$$-x^2(\log F_a)'' = r^2 - (a - 1 - d)r - a.$$

We prove positivity through an inequality between a sum of two nonnegative variables and their maximum.

Write $b = -\log x$, $z = \tanh b$, and $A = 1 + z$. Then $b, z \geq 0$ and $d = 1 - z$. For $s \geq 0$ define

$$h(s) = \frac{\tanh(b+s) - z}{A}, \qquad W = \int_0^\infty e^{-as}h'(s)\,ds, \qquad J = \int_0^\infty e^{-as}h(s)h'(s)\,ds.$$

In the defining integral of F_a, use $t = e^{-s}$ and $1/(1 + e^{-2(b+s)}) = (1 + \tanh(b + s))/2$. An integration by parts and the identity $1+x^2 = 2/A$ now yield $(1+x^2)F_a = 1+W$. Also $h'(0) = d$ and $-h'' = 2(z + Ah)h'$. Integrating the latter identity against e^{-as} gives

$$d - aW = 2zW + 2AJ.$$

The derivatives of h are bounded and decay exponentially at infinity, so the integrations are valid for every $a > 0$. Since $h(0) = 0$ and $h'(s) > 0$,

$$\begin{aligned} W^2 &= \int_0^\infty \int_0^\infty e^{-a(u+v)}h'(u)h'(v)\,du\,dv \\ &< \int_0^\infty \int_0^\infty e^{-a\max(u,v)}h'(u)h'(v)\,du\,dv = 2J. \end{aligned}$$

For the equality on the right, split the quadrant along its diagonal. As $z \geq 0$ and $A \geq 1$ this implies

$$d - (a + z)W - W^2 = zW + 2AJ - W^2 > 0.$$

Finally $r = a/(1 + W)$, and direct substitution gives

$$\frac{(1+W)^2}{a}\big(r^2 - (a - 1 - d)r - a\big) = d - (a + z)W - W^2.$$

This proves the strict inequality, including at $x = 1$. □

Corollary 12.2. *The point* 1 *is the largest possible common right endpoint for all the integer-indexed remainders. At every fixed* $x > 1$ *the logarithmic second derivative of* $T_{2n}[\arctan x]$ *is positive for all sufficiently large* n*.*

Proof. The probability measures $at^{a-1}dt$ on $[0, 1]$ converge weakly to the point mass at 1 as $a \to \infty$. For fixed $x > 0$ the kernel $(1 + x^2t^2)^{-1}$ and its first two derivatives in x are continuous on $0 \leq t \leq 1$. Therefore $(\log F_a)''(x)$ converges to $2(x^2 - 1)/(1 + x^2)^2$. Taking $a = 2n + 1$ gives the assertion. □

Source scope. Liu–Qi's monotonicity result for the same real remainder gives a sign for the first derivative. Their logarithmic convexity result for the imaginary-argument continuation concerns the normalized hyperbolic arctangent remainder. The present assertion concerns the second logarithmic derivative of the real arctangent remainder.

13 Derivative signs of exponential remainders at every real center

Y. Shuang, C.-Y. He, and F. Qi, J. Math. Inequal. **19** (2025), 1239–1251, doi:10.7153/jmi-2025-19-80, combine two assertions in Guess 3. Write $M_n(z) = T_n[e^z]$ for their normalized remainder. They ask whether M_n is logarithmically absolutely monotone on $\mathbb{R}$ and whether M_n/M_{n+1} is absolutely monotone there, for all integers $n \geq 0$. Here absolute monotonicity requires nonnegative derivatives of all orders, and logarithmic absolute monotonicity imposes this condition on derivatives of $\log M_n$ of orders at least one. These assertions subsequently appear separately in F. Qi, *Theory of Normalized Remainders in Taylor Series Expansions*, arXiv:2512.15797, Sections 12.7.1.3–4. We retain the original joint count.

Theorem 13.1. *Fix any real x and integer $n \geq 0$. For each of $g = \log M_n$ and $g = M_n/M_{n+1}$, both $\{k \geq 1 : g^{(k)}(x) > 0\}$ and $\{k \geq 1 : g^{(k)}(x) < 0\}$ have positive lower natural density. In particular both assertions of Guess 3 fail at every order parameter and every real center.*

Proof. Taylor's integral remainder and a change of variables give

$$M_n(z) = \frac{(n+1)!}{z^{n+1}} \left(e^z - \sum_{j=0}^{n} \frac{z^j}{j!} \right) = (n+1) \int_0^1 e^{zv} (1-v)^n \, dv.$$

Thus $M_n(0) = 1$, M_n is entire of order at most one, and it is positive on the real line. It has a complex zero. Otherwise Hadamard factorization would give $M_n(z) = e^{az+b}$, whereas $(\log M_n)''(0)$ is the strictly positive variance of the nondegenerate probability measure $(n+1)(1-v)^n \, dv$. All zeros are simple. Indeed, at a zero $\zeta \neq 0$ the derivative of the expression in parentheses is $\zeta^n/n!$, so $M_n'(\zeta) = (n+1)/\zeta$.

Consider either of the real meromorphic functions

$$f = M_n'/M_n, \qquad f = M_n/M_{n+1} = z/(n+2) + 1/M_{n+1}.$$

Each has a nonempty discrete set of simple nonreal poles. For the center x, write R for the distance to the nearest poles and $x + Re^{i\theta_j}$ for all poles at that distance. Writing c_j for their nonzero residues, choose $r > R$ which still excludes every farther pole, including its boundary. Subtract these principal parts and use Cauchy's coefficient estimate on the circle of radius r. For $k \geq 0$ this gives

$$\frac{R^{k+1}}{k!} f^{(k)}(x) = v_k + O((R/r)^k), \qquad v_k = -\sum_j c_j e^{-i(k+1)\theta_j}.$$

The sequence v_k is real by conjugation. Its arithmetic mean tends to zero because no θ_j is zero modulo 2π. Its mean square tends to $V = \sum_j |c_j|^2 > 0$ by the geometric-sum identity for distinct frequencies. Since $|v_k| \leq L = \sum_j |c_j|$, the averages of $|v_k|$ have lower limit at least V/L. The positive and negative parts therefore both have mean lower limit at least $V/(2L)$. Taking $\delta = V/(4L)$ shows that each of $v_k > \delta$ and $v_k < -\delta$ has lower density at least $V/(4L^2)$. The error in the coefficient estimate tends to zero. Both sign sets for the derivatives of f thus have positive lower density. For $f = M_n'/M_n$ these are precisely derivatives of $\log M_n$ with the index shifted by one, which preserves lower density. □

The argument uses classical factorization and coefficient estimates. The positive-density conclusion quantifies the failure of the original conjecture for every center and order parameter, with density bounds that may depend on those choices. Nearest-singularity explanations of high-cumulant oscillations have classical precedents, including C. Flindt et al., *Universal oscillations in counting statistics*, doi:10.1073/pnas.0901002106. The application here simultaneously treats all the exponential remainders, all their adjacent quotients, and every real center.

14 Adjacent moment tails for inverse sine and inverse tangent

The Eighth and Ninth Conjectures in Section 12.7.1 of Qi's chapter ask about the ratio $T_{2n}[f]/T_{2n+2}[f]$ for $n \geq 0$. They ask for concavity on $(-1,1)$ for $f = \arcsin$, and for increase on $(0,\infty)$ for $f = \arctan$. The following auxiliary problem treats both signs of the argument for one measure. It also gives an all-orders extension of the second assertion.

Let ν be a probability measure on $[0,1]$, with $m = \int s\,d\nu > 0$. Define its size-biased law $d\nu_1(s) = s\,d\nu(s)/m$, let $m_1 = \int s\,d\nu_1$, and put

$$F(z) = \int \frac{d\nu(s)}{1-zs}, \qquad F_1(z) = \int \frac{d\nu_1(s)}{1-zs}, \qquad R(z) = F(z)/F_1(z).$$

Proposition 14.1. *For some finite positive measure ρ on $[0,1]$,*

$$R(z) = 1 - (m_1 - m)z - z^2 \int \frac{d\rho(s)}{1-zs}, \qquad z < 1.$$

If ν is nondegenerate, $R(x^2)$ is strictly concave on $(-1,1)$. The function $w \mapsto R(-w)$ is then positive and strictly increasing for $w \geq 0$, with a completely monotone first derivative.

Proof. In $L^2(\nu_1)$ let A be multiplication by s, let $e = 1$, let Q project orthogonally onto $e^\perp$, and write $C = QA|_{e^\perp}$, $v = QAe$. Since $0 \leq C \leq I$, the Schur complement of $I - zC$ in $I - zA$ gives

$$F_1(z)^{-1} = 1 - zm_1 - z^2\langle v, (I - zC)^{-1}v\rangle.$$

The last inner product is the transform of the spectral measure of v for C. The identity $F = 1 + mzF_1$ proves the representation. This uses only the classical resolvent identity and spectral theorem.

Since $m_1 - m = \mathrm{Var}_\nu(s)/m$, the coefficient of z is strictly negative for nondegenerate ν. Every coefficient of higher degree is nonpositive, proving the strict concavity after setting $z = x^2$. For $w \geq 0$, $R(-w) = m/\mathbb{E}_w[s]$, where the probability in $\mathbb{E}_w$ has density proportional to $(1+ws)^{-1}$ relative to ν. Differentiation gives

$$\frac{d}{dw}\mathbb{E}_w[s] = -\operatorname{Cov}_w\left(s, \frac{s}{1+ws}\right) < 0,$$

because both functions of s are strictly increasing. For $j \geq 2$ the representation gives

$$(-1)^{j-1}\frac{d^j}{dw^j}R(-w) = j!\int \frac{s^{j-2}}{(1+ws)^{j+1}}\,d\rho(s) \geq 0.$$

This proves the derivative signs on the entire positive half-line. □

For arcsine the coefficients of $\arcsin x/x$ in x^2 are $a_j = 4^{-j}\binom{2j}{j}/(2j+1)$. These are the moments of $S = VU^2$, where U is uniform on $[0,1]$ and V is independent with the arcsine law $dv/(\pi\sqrt{v(1-v)})$. Use as ν the law of S weighted by S^n/a_n. Then $T_{2n}[\arcsin x] = F(x^2)$, and advancing n by one replaces F by F_1. Proposition 14.1 proves the Eighth Conjecture. For arctangent use $S = U^2$, whose moments are $(2j+1)^{-1}$. Then $T_{2n}[\arctan x] = F(-x^2)$. The identity holds on the full real line by the geometric remainder under the integral for $\arctan x/x$. It gives the strictly increasing quotient in the Ninth Conjecture for every $x > 0$.

Comparison with earlier results. For the Ninth Conjecture the hypergeometric representation is $T_{2n}[\arctan x] = {}_2F_1(1, n+1/2; n+3/2; -x^2)$. Karp and Sitnik, *Inequalities and monotonicity of ratios for generalized hypergeometric function*, J. Approx. Theory 161 (2009), 337–352, Theorem 1, already prove the required monotonicity on taking the upper and lower parameter shifts equal to one and inverting the ratio. For the Eighth Conjecture, the arcsine moment coefficients are strictly log-convex. Kaluza's classical reciprocal-series criterion gives nonpositive nonconstant coefficients of $1/F_1$. The identity $F/F_1 = mz + 1/F_1$ and $m_1 > m$ then prove the claimed concavity directly. These two items are excluded from the ten in Section 1.

15 Maximum-degree neighborhoods and Graffiti.pc Conjecture 165

Let G be a finite simple connected graph. Write $L(G)$ for the maximum number of leaves in a spanning tree, M for its set of maximum-degree vertices, and $t(G) = |N_G(M) \setminus M|$ for the number of distinct vertices outside M adjacent to it. The 8 August 2005 entry for Graffiti.pc Conjecture 165 asks whether $L(G) \geq \lceil \sqrt{2t(G)} \rceil$ [1]. The entry discusses a summer project of Craig Foster which asks for a universal constant c with $L(G) \geq c\, t(G)$. His examples require $c \leq 1/4$.

Theorem 15.1. *Every finite simple connected graph of maximum degree at least three satisfies*

$$t(G) \leq 4(L(G) - 2).$$

The coefficient 4 *is sharp. Conjecture 165 holds on its entire original domain, and the largest constant in Foster's multiplicative inequality is* $1/4$.

A component problem. Retain exactly the edges incident with M and discard isolated vertices. In each resulting component the vertices outside M are independent. The quantity $L - 2$ gives a budget for assembling solutions on these components, since for every tree T on at least two vertices,

$$\#\text{leaves}(T) - 2 = \sum_{u \in V(T)} (\deg_T u - 2)_+, \qquad x_+ = \max(x, 0). \tag{15.1}$$

In particular, extending vertex-disjoint trees to a spanning tree of a connected graph preserves each selected vertex's contribution to the right-hand side or increases it. We quantify how large the external part of a single component can be under this budget.

Lemma 15.2. *Let H be a finite simple connected graph with vertex partition $A \cup B$, where A is nonempty, B is independent, all vertices of A have a common degree $D \geq 3$, and the degrees in B are less than D. Then $|B| \leq 4(L(H) - 2)$.*

Proof. Set $h = |A|$, $t = |B|$, q equal to the number of components of $H[A]$, and $g = L(H) - 2$. The full star about a vertex in A extends to a spanning tree without decreasing the contribution of its center in (15.1), so $g \geq D - 2 \geq 1$. If H is a tree, (15.1) gives $g \geq h(D - 2)$. Also $t \leq Dh$ by counting neighbors, and $D \leq 3(D - 2)$, proving $t \leq 3g$ in this case.

Suppose H contains a cycle. Extend a spanning forest of $H[A]$ to a spanning tree T of H. All $h + t - 1$ edges of T touch A, and the $h - q$ edges in its prescribed forest touch A twice. Thus

$$g \;\geq \sum_{u \in A} (\deg_T u - 2) \;\geq\; (h + t - 1) + (h - q) - 2h = t - q - 1.$$

This gives $t \leq g + q + 1$. Every degree-two vertex belongs to B, so there is no pair of adjacent degree-two vertices. Bankevich and Karpov's Theorem 2 [2], applied with girth lower bound 3 and maximum degree-two chain length 1, gives

$$L(H) \geq (h + t - 3)/6 + 2.$$

Here its coefficient is $(3 - 2)/((3 - 1)(1 + 2)) = 1/6$. Consequently

$$t \leq g + 1 + q \leq g + 1 + h, \qquad h + t \leq 6g + 3, \qquad 2t \leq 7g + 4. \tag{15.2}$$

For $g \geq 4$ this implies $t \leq 4g$, while $g = 3$ gives $t \leq 12$ by integrality.

For the two remaining values we record a packing observation. Two nonadjacent degree-D vertices with disjoint neighborhoods give two disjoint full stars. Extending their union to a

spanning tree and using (15.1) yields at least $2D-2$ leaves. If the two centers have $s \geq 1$ common neighbors, join the centers by one of these neighbors and attach each remaining vertex in the union of their neighborhoods to one incident center. When both centers have other assigned neighbors, this tree has $2D-s-1$ leaves. Such an assignment exists since $D \geq 3$: if the two neighborhoods are identical, give each center a different remaining neighbor; otherwise each has a private neighbor. Extending the tree gives $L(H) \geq 2D-s-1$.

If $g=1$ and $t>4$, (15.2) forces $t=5$ and $q \geq 3$. We have $D=3$. Take one center in each of three distinct components of $H[A]$. Every pair of these nonadjacent centers must have at least two common neighbors by the packing observation and $L(H)=3$. A common neighbor lies in B, since otherwise the centers belong to the same component of $H[A]$. Each vertex of B has degree at most two and serves at most one pair. The three pairs thus require six distinct vertices of B, contradicting $t=5$.

If $g=2$ and $t>8$, (15.2) gives $t=9$ and $6 \leq q \leq h \leq 6$. Thus A is independent and $D \in \{3,4\}$. When $D=4$, the packing observation and $L(H)=4$ require at least three common neighbors for every pair of centers. There are fifteen pairs. On the other hand,

$$\sum_{b \in B} \binom{\deg_H b}{2} \leq \sum_{b \in B} \deg_H b = 24,$$

since $\deg_H b \leq 3$. This contradicts the required count of at least 45. For $D=3$ the degree sum over B is 18, and each of its nine vertices therefore has degree exactly two. Suppress them to obtain a connected loopless cubic multigraph on the six centers with nine edges. Take a spanning tree of the multigraph and fix a leaf a in it. Subdivide each of its five edges as in H. For each of the four unused edges, attach its subdivision vertex as a leaf to an endpoint different from a. Such an endpoint exists because the multigraph is loopless. The resulting spanning tree of H has those four leaves and a, contradicting $L(H)=4$. This completes the lemma. □

Proof of Theorem 15.1. Let the maximum degree of G be $D \geq 3$. Form the spanning subgraph with all edges incident to M. Its nontrivial components H satisfy Lemma 15.2 with $A = V(H) \cap M$ and $B = V(H) \setminus M$. Their B sets partition $N_G(M) \setminus M$. In each H choose a spanning tree with $L(H)$ leaves. The trees form a forest in G, and extend to a spanning tree of G. Equation (15.1) gives

$$t(G) = \sum_{H} |V(H) \setminus M| \leq 4 \sum_{H} (L(H)-2) \leq 4(L(G)-2).$$

In particular $L(G)^2 - 2t(G) \geq (L(G)-4)^2 \geq 0$. For maximum degree at most two the possibilities are a singleton, a path and a cycle. The singleton, K_2, and every cycle have $t=0$. A path on at least three vertices has $t=2$ and $L=2$. These cases also satisfy the stated square-root inequality and $L \geq t/4$.

For sharpness, the original annotation credits Foster with the parameters $L=k+2$, $t=4k$. An explicit realization described in VMRC [066] is as follows. For each cyclic index $i=1,\dots,k$ take two length-two paths from a_i to b_i and one length-three path from b_i to a_{i+1}, with all internal vertices distinct. The $2k$ endpoints have degree three. The $4k$ internal vertices comprise their external neighborhood. Every spanning tree breaks each of the k vertex-disjoint four-cycles, so at most one endpoint of each pair retains degree three. Thus $L \leq k+2$ by (15.1), and the bound just proved gives equality. Consequently L/t tends to $1/4$, as required. □

Historical scope. The 2006 UHD conference abstract by Foster [3] reports the resolution of several Graffiti.pc conjectures and discusses this specific pair of invariants. It does not identify those conjecture numbers or state the inequalities and hypotheses. Its scope therefore leaves a substantive priority question. Corpus [066] treats the case where M is independent,

$D \in \{3,4,5\}$ and the remaining degrees are at most two. The proof above gives the result for arbitrary connected graphs. Historical priority for the full inequality or its method has not been established.

16 A sharp two-shift classification for the Qi–Luo question

Qi's 2011 preprint (arXiv:1104.4442, Remark 5) and the revised article by F. Qi and Q.-M. Luo, *Complete monotonicity of a function involving the gamma function and applications*, Period. Math. Hungar. **69** (2014), 159–169, doi:10.1007/s10998-014-0056-x, Remark 4.6, ask for the logarithmic complete monotonicity of their two-parameter function (4.5). For a correction amplitude $c > 0$ put

$$G_{\lambda,\mu}^{(c)}(x) = \frac{e^x \Gamma(x+1)}{(x+\mu)^{x+\mu}} \exp\!\left(\frac{c}{x+\lambda}\right), \qquad x > \max(0, -\lambda, -\mu).$$

Both shifts are arbitrary real numbers. The source fixes $c = 1/24$. We ask in addition which correction amplitudes allow the region of logarithmic complete monotonicity to be read from the two endpoint rates of a Laplace kernel.

Theorem 16.1. *For every $c \geq 1/24$ and all real λ, μ, the function $G_{\lambda,\mu}^{(c)}$ is logarithmically completely monotone on its full domain if and only if*

$$\mu \geq \max\{1/2, \min(1, \lambda)\}.$$

The number $1/24$ is the smallest positive amplitude for which this criterion holds for every pair of shifts. Taking $c = 1/24$ answers the two-parameter question in Remark 4.6.

Proof. Logarithmic complete monotonicity is equivalent to complete monotonicity of the negative logarithmic derivative. The latter is

$$\log(x+\mu) - \psi(x+1) + \frac{c}{(x+\lambda)^2} = \int_0^\infty e^{-xt} \frac{K(t)}{t}\, dt, \quad K(t) = \frac{t}{e^t - 1} - e^{-\mu t} + ct^2 e^{-\lambda t}.$$

This follows by subtracting the integral for the digamma function from Frullani's formula for the logarithm and adding the elementary Laplace integral for the last term. The density is bounded near zero and integrable after multiplying by e^{-xt} for every admissible x. Its continuity, Bernstein's representation theorem, and uniqueness of the Laplace transform after a fixed admissible exponential tilt give the criterion $K(t) \geq 0$ for every $t > 0$.

Since $K(t) = (\mu - 1/2)t + O(t^2)$, necessity gives $\mu \geq 1/2$. If $\mu < \min(1, \lambda)$, multiplying $K(t)$ by $e^{\mu t}$ gives limit -1 at infinity. These are the two necessary endpoint conditions.

For sufficiency it is enough to treat $c = 1/24$ and the smallest allowed μ, since increasing either only increases K. For $u > 0$,

$$\frac{u}{\sinh u} + \frac{u^2}{6} \geq 1.$$

Indeed the series of $(1 - u^2/6)\sinh u$ has coefficient one at degree one and coefficient $[1 - (2j)(2j+1)/6]/(2j+1)! \leq 0$ at degree $2j+1$ for $j \geq 1$. Put $u = t/2$, so $e^{t/2}t/(e^t - 1) = u/\sinh u$. For $\lambda \leq 1/2$ take $\mu = 1/2$, and observe

$$e^{\mu t} K(t) = \frac{u}{\sinh u} - 1 + e^{(1/2-\lambda)t} \frac{u^2}{6} \geq 0.$$

For $1/2 \leq \lambda \leq 1$ take $\mu = \lambda$, giving

$$e^{\mu t} K(t) = e^{(\lambda - 1/2)t} \frac{u}{\sinh u} - 1 + \frac{u^2}{6} \geq 0.$$

For $\lambda \geq 1$ take $\mu = 1$. Here $e^t t/(e^t - 1) = t/(1 - e^{-t}) > 1$. Finally, at $\lambda = \mu = 1/2$ the normalized kernel is $u/\sinh u - 1 + ct^2 = (c - 1/24)t^2 + O(t^4)$, so every smaller amplitude fails for this pair. □

The reciprocal. Although Remark 4.6 asks explicitly about G itself, its reciprocal at the source amplitude has the following sharp variational threshold. It is logarithmically completely monotone if and only if $\mu \le 1/2$ and

$$\lambda \ge \Lambda(\mu) := \max_{t>0} \frac{1}{t} \log \frac{t^2}{24(e^{-\mu t} - t/(e^t - 1))}.$$

Indeed the requisite kernel inequality is exactly $K(t) \le 0$. The expansion at zero requires $\mu \le 1/2$. Under that condition the parenthesized difference is positive because $u/\sinh u < 1$. Taking logarithms gives the formula with a supremum. The expression in the maximum tends to $-\infty$ at zero if $\mu < 1/2$, and to $1/2$ there if $\mu = 1/2$. At infinity it is $\mu + (2\log t - \log 24 + o(1))/t$, so it tends to μ from above. These endpoint facts imply the maximum is attained at a finite positive point. Qi's one-shift threshold at $\mu = 1/2$ is a predecessor of this formula. The boundary in Theorem 16.1, which answers the source's direct question, is explicit.

The proof uses classical Laplace positivity and an elementary hyperbolic inequality. The original source studies the fixed shift $\mu = 1/2$ with $\lambda \ge 0$, and cites the classification without the exponential correction. A bounded search did not locate the full two-shift result. Its historical priority has not been established.

17 Evidence and remaining questions

For the two-data determinant, the earlier diffusion-wave result of Floridia and Yamamoto, *Backward problems in time for fractional diffusion-wave equation*, arXiv:2007.09364, concerns the negative real ray. The rotating complex ray of Theorem 3.1 is the one in the original Schrödinger problem. A search of cited and citing works and variants of the determinant did not locate the same disproof for this ray. The same authors still discuss the complete-monotonicity question in their subsequent work arXiv:2603.16726. The contour expansion, winding number, and scale mixtures used here have classical origins. The quantitative bounds and transfers are stated separately so their novelty can be assessed directly.

The graph argument was checked separately against the degree-two chain hypothesis in the Bankevich–Karpov theorem and against the small component cases. The 2006 Foster abstract is a material gap in the priority assessment. It discusses the same invariants but does not identify all its results. A published source for the full linear bound was not located.

For the normalized gamma-ratio derivative the argument uses Kristiansen's theorem on gamma mixtures of shape two. Earlier work of Hansen [7] also relates canonical convolution measures to logarithmic curvature. The particular density comparison and the full application were checked against the source formulas and against a published statement giving an incompatible answer. The explicit gap $r = 2$ makes that disagreement verifiable independently. For Qi–Wei–Guo, the work before the journal article dates to 2009. A bounded search of citing works did not locate the sharp reciprocal threshold. The Qi–Luo proof uses a classical Laplace representation and a short hyperbolic inequality. These comparisons do not certify a historically new general principle in every case.

The fractional-zero proofs combine analytical estimates with rigorously rounded interval calculations. The exact statement with coefficient one is distinct from the fitted constants in Conjectures 1 and 3. Hanneken, Achar and Vaught [6] had already computed a close numerical value for the critical order. The results above certify its location, establish positivity at all times below it and describe both branches of the bifurcation locally. The implicit-function and subordination inputs are classical. The uniform two-parameter estimate is stated with additive error so its second-order terms can be tested. Independent internal audits checked the finite and infinite portions of the arguments. External peer review and exhaustive priority verification are pending.

All four interval-verification programs are embedded in the single TeX source after `\end{document}`. Copy each block delimited by its matching `BEGIN` and `END` comments into its indicated Python file and run it with Python-FLINT, which supplies Arb ball arithmetic. No external figure, additional TeX file, or bibliography file is needed for the PDF.

A Typical shifts in the smooth-component question of Erdős and Graham

For an integer $t \geq 3$, let $s_t(m) = \prod_{p<t} p^{v_p(m)}$ for positive integers m. Write $f(n,t)$ for the number of distinct $s_t(n+i)$ as $1 \leq i \leq t$. Erdős and Graham's question (problem 461 in the Erdős Problems database, from *Old and New Problems and Results in Combinatorial Number Theory*, 1980, p. 92) asks for a positive lower bound on $f(n,t)/t$ *uniformly* in n and t. We give an asymptotic valid in mean square over the shifts. The uniform question remains open in this work.

Let $H_j = \sum_{a=1}^{j} 1/a$, $V_t = \prod_{p<t}(1-1/p)$, and $M_t = \prod_{p<t} p^{\lceil \log t/\log p \rceil}$. Let $\mathbb{E}_t$ denote the uniform average over a full period of M_t consecutive shifts.

Theorem A.1. *As $t \to \infty$,*

$$\mathbb{E}_t \left(\frac{f(n,t)}{t} - (1 - V_t H_{t-1}) \right)^2 = O\left(\frac{\log\log t}{\log t} \right).$$

Consequently $f(n,t)/t$ converges in mean square, and in natural-density probability over the shifts, to $1 - e^{-\gamma}$.

Proof. Every label $s_t(n+i) \geq t$ occurs once in the window, since equal labels divide the difference of their positions. The events that the label is d for any $d < t$, or is at least t, are determined modulo M_t. Indeed, reducing each valuation to the exponent in M_t either preserves the label or leaves a factor at least t. We may therefore choose one independent uniform p-adic shift N_p for each $p < t$. The statistics used for the exact window count have precisely the full-period law by the Chinese remainder theorem. Valuations are almost surely finite. At one position let $S = \prod_{p<t} p^{v_p(N_p+i)}$. For any t-smooth integer d (all prime factors strictly below t), independence and the geometric valuation law give

$$\Pr(S = d) = \frac{V_t}{d}.$$

In particular, with $Z_i = \mathbf{1}_{\{S_i \geq t\}}$, $\mathbb{E}Z_i = 1 - V_t H_{t-1}$.

Take $y = (\log t)^2$. At one position split $S = UX$ into its prime powers for $p \leq y$ and $y < p < t$, respectively. The factors U, X are independent. Put $V_{y,t} = \prod_{y<p<t}(1-1/p)$. Conditional on $U = m$, the chance that dropping U changes the indicator of $S \geq t$ is at most

$$V_{y,t} \sum_{\substack{\max(1,t/m) \leq d < t \\ P^-(d) > y}} \frac{1}{d},$$

where $P^-(1) = \infty$. We use a harmonic packing bound. Multiply each such d by every integer $1 \leq b \leq y$. The products bd are distinct, since b is their entire component supported on primes up to y. All the products lie between $\max(1, t/m)$ and ty. Comparing their harmonic sum with the harmonic sum over that whole interval gives

$$H_{\lfloor y \rfloor} \sum_{\substack{\max(1,t/m) \leq d < t \\ P^-(d) > y}} \frac{1}{d} \leq 1 + \log m + \log y.$$

The lost-indicator probability is at most

$$q_y = \frac{V_{y,t}}{H_{\lfloor y \rfloor}} \left(1 + \log y + \sum_{p \le y} \frac{\log p}{p-1} \right) = (2 + o(1)) \frac{\log y}{\log t}.$$

Here Mertens' two prime sum and product estimates apply whenever $y = y(t) \to \infty$ and $y < t$. The estimate is independent of the position.

For two positions distance $1 \le h < t$ apart, compare their paired vectors of valuations for $p > y$ with two independent vectors having the correct marginals. If $p \nmid h$, simultaneous positive valuations are impossible. Comparing the probabilities of the zero-zero cell, the two axes, and the positive-positive cells gives total variation distance $2/p^2$. If $p \mid h$, the distance is at most $4/p$ by comparison to the point mass at $(0, 0)$. The prime-wise pairs are independent. Coupling each pair separately and using a union bound gives total variation distance at most

$$2 \sum_{p > y} p^{-2} + 4 \sum_{\substack{p > y \\ p \mid h}} p^{-1} \ll \frac{1}{y} + \frac{\log t}{y \log y}.$$

The last estimate follows since $h < t$ has at most $(\log t)/(\log y)$ distinct prime divisors larger than y. The covariance of the indicators $\mathbf{1}_{\{X_i \ge t\}}$, $\mathbf{1}_{\{X_j \ge t\}}$ is bounded in absolute value by this total variation distance. Restoring the small primes changes that covariance by at most $4q_y$. Consequently

$$\operatorname{Var} \left(t^{-1} \sum_{i=1}^{t} Z_i \right) \ll t^{-1} + q_y + y^{-1} + \frac{\log t}{y \log y} \ll \frac{\log \log t}{\log t}.$$

It remains to count distinct labels smaller than t. Write D for their number and take $K = \lfloor t / \log t \rfloor$. There are at most K possible labels up to K. The number of distinct labels in (K, t) is at most the number of their occurrences, so

$$\frac{\mathbb{E}D}{t} \le \frac{K}{t} + V_t \sum_{K < d < t} \frac{1}{d} \ll \frac{\log \log t}{\log t}.$$

Since $0 \le D \le t$, the same upper bound holds for $\mathbb{E}[(D/t)^2]$. The exact identity $f = \sum_i Z_i + D$ and $(a+b)^2 \le 2a^2 + 2b^2$ give the claimed estimate. Finally $V_t H_{t-1} \to e^{-\gamma}$ by Mertens' product theorem. □

The publicly recorded averaging argument on the discussion page for problem 461 obtains the upper bound $\inf_n f(n,t)/t \le 1 - e^{-\gamma} + o(1)$. The pointwise marginal used above is also the one relevant to that bound. For the original one-sided question a concentration inequality gives a stronger exceptional-set estimate.

Corollary A.2. *Fix $0 < c < 1 - e^{-\gamma}$ and $0 < a < (1 - e^{-\gamma} - c)/2$. There is $C_{a,c} > 0$ such that for all sufficiently large t,*

$$\Pr_t \{ f(n,t) < ct \} \le \exp(-C_{a,c} t^a).$$

The estimate holds after fixing arbitrary residue classes of the shift at all prime powers in M_t with base $p \le t^a$ and averaging only over the remaining coordinates.

Proof. Use $y = t^a$. At each position the indicator of $X_i \ge t$ depends only on the remaining prime coordinates. The number B of positions where this happens is a lower bound for f for *every* choice of the small-prime residues. Indeed each such position must have full smooth label at least t, and such labels are distinct. The packing bound in the preceding proof yields

$$\mathbb{E}B/t \;\ge\; 1 - V_t H_{t-1} - q_y \;\ge\; 1 - e^{-\gamma} - 2a - o(1) > c + \delta$$

for some $\delta > 0$. Changing the shift coordinate at one $p > y$ affects at most $2\lceil t/p \rceil$ entries, since the affected positions are confined to the old and new zero residue classes modulo p. Thus the sum of squares of coordinate oscillations of B/t is at most

$$\sum_{y<p<t} (2\lceil t/p \rceil/t)^2 \le 16 \sum_{m>y} m^{-2} \ll 1/y.$$

The bounded-differences inequality for independent coordinates gives $\Pr(B/t < c) \le \exp(-C_{a,c}y)$, as claimed. This inequality follows, for example, by revealing the prime coordinates one at a time and applying Hoeffding's exponential bound conditionally to the martingale increments of $\mathbb{E}[B/t \mid \text{revealed coordinates}]$. □

Corollary A.3. *Let $c_0 = 1 - e^{-\gamma}$ and fix $\epsilon > 0$ and $0 < a < \min(1, \epsilon/3)$. Uniformly after fixing arbitrary residue classes at all prime powers in M_t with base $p \le t^a$, one has*

$$\Pr_t\{|f(n,t)/t - c_0| > \epsilon\} \le 2\exp(-C_{a,\epsilon}t^a)$$

for sufficiently large t. In particular, the periodic extension of f to a Haar-random profinite integer N satisfies $f(N,t)/t \to c_0$ almost surely through all integer t.

Proof. Keep $y = t^a$, with the coordinates at primes up to y now fixed. Let U_i be the product of their prime powers at position i, with each valuation capped at its exponent e_p in M_t. If $U_i < t$ no cap is effective. If $U_i \ge t$ the full label is necessarily a singleton, independently of every higher valuation. For these arbitrary fixed residues, counting multiples yields

$$\frac{1}{t}\sum_{i=1}^{t} \log\min(U_i, t) \le \sum_{p\le y} \log p \sum_{k=1}^{e_p} \left(p^{-k} + t^{-1}\right) \le \log y + O(1) + O(y\log t/t).$$

Indeed $e_p \log p \le \log t + \log p \le 2\log t$. To obtain a conditional upper bound on the expected singleton count $\sum_i Z_i$, compare it with B. At position i a gain over B requires $t/\min(U_i, t) \le X_i < t$. The harmonic packing estimate with $m = \min(U_i, t)$ bounds the probability by

$$\frac{V_{y,t}}{H_{\lfloor y \rfloor}}(1 + \log y + \log\min(U_i, t)).$$

Its average over i is at most $2a + o(1)$ uniformly in the fixed coordinates. Also $\mathbb{E}B/t \le 1 - V_t H_{t-1}$, by first reintroducing independent small-prime coordinates in the unrestricted calculation. Consequently the conditional expected count $t^{-1}\sum_i Z_i$ is between $c_0 - 2a - o(1)$ and $c_0 + 2a + o(1)$.

To bound the conditional expectation of D/t, again split at $K = \lfloor t/\log t \rfloor$. If $U_i \ge t$ there is no small label at this site. Otherwise the chance that the label belongs to (K, t) is $V_{y,t}$ times a reciprocal sum over some of the y-rough integers in $(K/U_i, t/U_i)$. Multiplying each term by all $b \le y$, as above, bounds the latter sum by

$$\frac{1 + \log(yt/K)}{H_{\lfloor y \rfloor}}.$$

This works also when $K/U_i < 1$, since the lower endpoint of the harmonic sum after multiplication is then at least one. Adding K/t gives $\mathbb{E}[D/t \mid \text{fixed coordinates}] \le a + o(1)$. Combining both bounds places the conditional expectation of f/t between $c_0 - 2a - o(1)$ and $c_0 + 3a + o(1)$, uniformly for the specified small-prime residues.

Changing the coordinate of one prime $p > y$ alters at most $2\lceil t/p \rceil$ sites. Changing m site labels in a multiset alters its number of distinct labels by at most m. One may use the truncated labels $\gcd(n + i, M_t)$ directly for this argument, since all labels at least t are singletons even after truncation. Thus the bounded-differences estimate from Corollary A.2 applies to f/t itself. This gives the asserted two-sided bound because $\epsilon > 3a$. Summing the resulting unrestricted upper bound over t and invoking the first Borel–Cantelli lemma for each positive rational ϵ proves the almost-sure claim. □

For comparison, each fixed ordinary shift $n \geq 0$ satisfies $f(n,t)/t \to 1$, since the $m < t$ in its window all equal their own smooth components. At $n = 0$ the number of labels at least t is at most one, although there are at least $t-1$ distinct labels. Thus the singleton estimate used in the concentration proof cannot give a uniform linear lower bound by itself. The natural density and fixed-shift limits have different quantifiers.

An exceptional set can have positive cardinality even when its proportion of the full period is exponentially small in a power of t. The estimate leaves the uniform lower-bound question unresolved.

B Free isospectral potentials with four occupied sites

Faust, Friedman, O'Malley, Ramos, and Sharma ask in Conjecture 2 of *Algebraic properties of the ideal of spectral invariants for the discrete Laplacian* (arXiv:2602.08304) for a nonzero complex q-periodic potential isospectral to the free one and satisfying a conjugate-reflection symmetry, for every $q > 3$. Faust et al., J. Math. Phys. **65** (2024), 073501, constructed such even-period potentials using four occupied sites. We determine exactly when a four-site construction is possible, allowing arbitrary complex values. The existence question for odd periods and unrestricted support remains open in this work.

For $t \neq 0$ and $x = t + t^{-1}$ write $[j] = (t^j - t^{-j})/(t - t^{-1})$. For positive j this is a monic polynomial of degree $j-1$ in x. The two-sided recurrence is $[j+1] = x[j] - [j-1]$, with $[0] = 0$ and $[1] = 1$. The discriminant is the trace of the ordered product of the matrices $T_j = \left(\begin{smallmatrix} x - v_j & -1 \\ 1 & 0 \end{smallmatrix}\right)$. Equality of the discriminants is equivalent to Floquet isospectrality since the transfer matrices have determinant one.

Theorem B.1. *Let $q \geq 4$. A nonzero free-isospectral complex potential supported on at most four sites has exactly four occupied sites. Up to cyclic choice of the initial occupied site, their consecutive values must be*

$$(A, -A, B, -B), \qquad B = i\epsilon A, \quad \epsilon \in \{-1, 1\}.$$

Write (a, b, c, d) for the consecutive positive cyclic gaps, starting with the edge from A to $-A$. Then $b = d = 1$, and exactly the following possibilities occur:

1. *$a = c \geq 1$, $q = 2a + 2$, $A^2 = 2i\epsilon$;*
2. *$\{a, c\} = \{1, 2\}$, $q = 5$, $A^2 = 2i\epsilon + \operatorname{sgn}(c - a)$.*

Conversely every listed pattern is free-isospectral. The second case does not have a conjugate-invariant multiset of entries. In particular, for every odd $q \geq 5$ any potential satisfying the original conjecture requires at least five occupied sites.

Proof. Expand the discriminant multilinearly using $T_j = T - v_j e_1 e_1^{\mathsf{T}}$, where T is the free transfer matrix. For any nonempty set I of k occupied sites, with successive cyclic gaps $g_1, \ldots, g_k$, the contribution to the difference from the free discriminant is

$$(-1)^k \prod_{j \in I} v_j \prod_{l=1}^{k} [g_l].$$

Indeed $e_1^{\mathsf{T}} T^{g-1} e_1 = [g]$, as follows from the recurrence. The product of gap polynomials has degree $q - k$, parity $q - k$, and leading coefficient one. The $k = 1$ group is $-[q] \sum v_j$, so the highest coefficient forces $e_1(v) = 0$ and this group vanishes identically. The next two coefficients therefore force $e_2(v) = e_3(v) = 0$ (at the third coefficient the $k = 2$ group contributes nothing by parity). With at most three occupied sites, these identities force every entry to be zero.

With four occupied sites, the four values are the roots of $X^4 - A^4$ for some $A \neq 0$, namely $A, iA, -A, -iA$.

The $k = 3$ group has opposite parity to the groups $k = 2, 4$, hence must vanish individually. Denote the four cyclic values by v_j, and the edge lengths from j to $j+1$ by g_j. Dividing the triple sum by $\prod_{j=1}^{4} v_j[g_j]$ gives

$$\sum_{j=1}^{4} v_j^{-1} \frac{[g_{j-1} + g_j]}{[g_{j-1}][g_j]} = 0.$$

The ratio $[u+v]/([u][v])$ equals $(t - t^{-1})(H_u + H_v)/2$, where $H_u = (t^{2u}+1)/(t^{2u}-1)$. The rational functions H_u for distinct positive u are linearly independent: a primitive $2u$-th root is a pole of the largest-index term and no smaller one. Iteration proves independence. Consequently for every occurring length g,

$$\sum_{j : g_j = g} (v_j^{-1} + v_{j+1}^{-1}) = 0. \qquad (*)$$

If no cyclically neighboring values are opposites, we may list the values as $(A, B, -A, -B)$ with $B = i\epsilon A$. In (*) the four edge weights consist of two pairs of opposites, represented by $A^{-1} + B^{-1}$ and $B^{-1} - A^{-1}$, which are linearly independent over the reals. Thus the cyclic gaps are (a, b, a, b). The residual from the pair and quartet groups is

$$2i\epsilon A^2([a][q-a] - [b][q-b]) - A^4[a]^2[b]^2.$$

Here $q = 2a + 2b$ and the difference in parentheses equals $[a-b][a+b] = [a]^2 - [b]^2$. Its vanishing would require $A^2 = 2i\epsilon([b]^{-2} - [a]^{-2})$ as a rational-function identity. If both $a, b > 1$ the right side tends to zero at infinity. If exactly one is one it varies with x, while if both are one it vanishes. This rules out this cyclic order.

Otherwise the cyclic values can be written $(A, -A, B, -B)$ with $B = i\epsilon A$. Formula (*) gives gaps (a, b, c, b). Directly summing the six pair terms and the quartet now gives, after division by $A^2[a][c]$,

$$\frac{[c-a][2b]}{[a][c]} + 2i\epsilon - A^2[b]^2 = 0. \qquad (**)$$

To verify this simplification, the opposite-pair terms before division are $A^2([c][a+2b] - [a][c+2b]) = A^2[c-a][2b]$, and $[a+b][b+c] - [b][a+b+c] = [a][c]$ for the two cross-pair lengths. These identities follow by expanding brackets in t. For real $x > 2$, take imaginary parts of (**). Since $2\epsilon \neq 0$, $[b]^2$ must be constant, giving $b = 1$. If $a = c$ this gives the first family. For $a \neq c$ the nonzero ratio $[c-a][2]/([a][c])$ has degree $2 - 2\min(a, c)$ at infinity, so it can only be constant if $\min(a, c) = 1$. With $u = \max(a, c) > 1$ its absolute sign-adjusted expression is $x[u-1]/[u]$. It tends to one at infinity, and the recurrence $x[u-1] = [u] + [u-2]$ shows it is identically one exactly when $u = 2$. This gives the second family. All steps after the vanishing of e_1, e_2, e_3 characterize the discriminant identity, proving the converse. In the second case $A^4 = -3 + 4i\epsilon \operatorname{sgn}(c-a)$ is nonreal, so the polynomial $X^4 - A^4$ is not real. Thus conjugation does not preserve the multiset of its roots. □

An independent support-width obstruction. If the shortest cyclic interval covering the support of a nonzero q-periodic free-isospectral potential contains h sites, then $h \geq \lfloor q/2 \rfloor + 2$. The even four-site family above attains this bound. To see this, place the occupied endpoints of the interval at $1, h$, write $M = T_h \cdots T_1$, and use the eigenbasis $S = \left(\begin{smallmatrix} t & t^{-1} \\ 1 & 1 \end{smallmatrix}\right)$ of T. Set $U(t) = (S^{-1}MS)_{11}$ and $P(t) = (t - t^{-1})(U(t) - t^h)$. The other diagonal entry is $U(1/t)$. Equality of the two period discriminants thus means $t^{q-h}P(t) = t^{h-q}P(t^{-1})$. On expanding the transfer product, the contribution of $j_1 < \cdots < j_k$ to $P(t)$ is

$$(-1)^k \prod v_{j_i}\, t^{h - j_k + j_1} \prod_{i=1}^{k-1} [j_{i+1} - j_i].$$

Its smallest exponent equals $h-2(j_k-j_1)+k-1$. For $h \geq 2$ the unique smallest exponent over all subsets is $3-h$, achieved only by the pair $\{1,h\}$, with nonzero coefficient $v_1 v_h$. Therefore the Laurent polynomial $t^{q-h}P(t)$ has lowest exponent $q-2h+3$. Since it equals its reciprocal and is nonzero, $q-2h+3 \leq 0$. This gives the stated bound. When $h=1$ the discriminant difference is simply $-v_1[q]$.

The 2024 construction already gives the even-period family and the 2026 source lists computed solutions for period five. Corpus [061] also gives many individual odd periods. The all-period four-site classification and width bound above address support restrictions. The original existence question at arbitrary odd periods with unrestricted support remains open in this work.

References

[1] E. DeLaViña, *Written on the Wall II*, Graffiti.pc Conjecture 165, 8 August 2005. Original catalogue. An archived copy of the HTML row with its Foster annotation is available in the Graffiti verification archive.

[2] A. Bankevich and D. Karpov, *Bounds of a number of leaves of spanning trees*, arXiv:1111.3266 (2011), Theorem 2. https://arxiv.org/abs/1111.3266.

[3] T. C. Foster, *On Leafy Spanning Trees of Graph*, 5th Annual Student Research Conference, University of Houston–Downtown (2006), conference abstract. https://www.uhd.edu/documents/academics/sciences/scholars/proceedings2006.pdf.

[4] R. Chaudhary, K. Diethelm and S. Hashemishahraki, *On the separation of solutions to fractional differential equations of order* $\alpha \in (1,2)$, Applied Numerical Mathematics 203 (2024), 84–96. doi:10.1016/j.apnum.2024.05.020. arXiv:2401.14771.

[5] R. Gorenflo and F. Mainardi, *Fractional calculus: integral and differential equations of fractional order*, in Fractals and Fractional Calculus in Continuum Mechanics (1997), 223–276. https://arxiv.org/abs/0805.3823.

[6] J. W. Hanneken, B. N. N. Achar and D. M. Vaught, *An alpha-beta phase diagram representation of the zeros and properties of the Mittag–Leffler function*, Advances in Mathematical Physics (2013), article 421685. doi:10.1155/2013/421685.

[7] B. G. Hansen, *On log-concave and log-convex infinitely divisible sequences and densities*, Annals of Probability 16 (1988), 1832–1839. doi:10.1214/aop/1176991600.

[8] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monographies de l'Enseignement Mathématique 28 (1980), p. 92. https://www.erdosproblems.com/461.